

A Project Report on

VidyaLok An AI-Powered Smart Library Ecosystem using NLP and Predictive Models

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

By

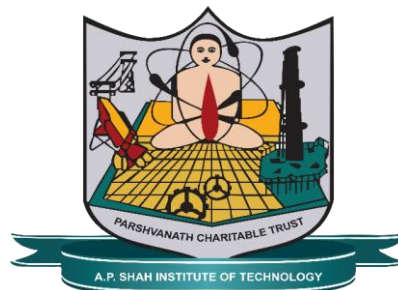
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2025-2026



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CERTIFICATE

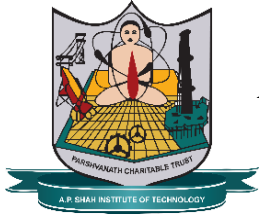
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Abstract

VidyaLok is an AI-powered smart library ecosystem designed to enhance traditional academic library systems by integrating intelligent, automated, and user-centric features. Conventional libraries often rely on manual processes, basic search mechanisms, and uniform services, which limit efficiency and student engagement. VidyaLok addresses these challenges through a unified platform that incorporates secure QR-based authentication, personalized book recommendation, semantic search, and real-time analytics. The system leverages machine learning and natural language processing techniques to improve resource discovery, reduce search time, and provide relevant academic materials based on user preferences, departmental context, and borrowing history.

In addition to improving user experience, the system introduces data-driven functionalities such as occupancy monitoring using entry logs, sentiment-based feedback analysis, and an analytics dashboard for administrators. These features help optimize resource utilization and enable informed decision-making. Built using modern web technologies and open-source tools, VidyaLok offers a scalable and cost-effective solution suitable for educational institutions. By transforming a traditional library into a smart and intelligent ecosystem, the project contributes to enhancing learning efficiency and operational management. The system aligns with Sustainable Development Goal 4 (Quality Education) by improving access to learning resources and Goal 9 (Industry, Innovation, and Infrastructure) by promoting digital innovation in institutional infrastructure.

Keywords: Smart Library Automation, QR Code Authentication, Cryptographic Authentication, Hybrid Recommendation Systems, Occupancy Monitoring, Real-time Analytics.

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ABBREVAION

AI	<i>Artificial Intelligence</i>
ML	<i>Machine Learning</i>
IoT	<i>Internet of Things</i>
RFID	<i>Radio Frequency Identification</i>
NLP	<i>Natural Language Processing</i>
QR	<i>Quick Response</i>
RAG	<i>Retrieval-Augmented Generation</i>
HMAC	<i>Hash-based Message Authentication Code</i>
BERT	<i>Bidirectional Encoder Representations from Transformers</i>
API	<i>Application Programming Interface</i>
SPLM	<i>Smart Public Library Management</i>
TF-IDF	<i>Term Frequency–Inverse Document Frequency</i>
PMI	<i>Pointwise Mutual Information</i>
SDN	<i>Software-Defined Networking</i>
GCN	<i>Graph Convolutional Network</i>
CF/CB	<i>Collaborative Filtering / Content-Based Filtering</i>

Chapter 1

Introduction

1. Introduction

Academic libraries are essential for supporting education and research, but traditional systems often rely on manual processes, static catalogs, and basic keyword-based search. These limitations reduce efficiency and fail to meet the expectations of modern students who are accustomed to personalized and intelligent digital platforms [1], [2]. Although technologies like RFID and cloud-based systems have improved automation and accessibility, they still lack advanced features such as personalization, predictive analytics, and intelligent interaction [3], [4], [6].

The emergence of Artificial Intelligence (AI) and Machine Learning (ML) has opened new opportunities to transform libraries into smart, user-centric systems. Recommendation systems, including collaborative and hybrid approaches, have shown significant improvements in resource discovery and user engagement [7], [9], [18]. Recent advancements further integrate deep learning and transformer-based models to enhance recommendation accuracy and adaptability [2], [12]. Additionally, Natural Language Processing (NLP) enables semantic search and automated summarization, allowing users to efficiently access and understand academic content [6], [21], [23].

Security and access control are also critical in modern library systems. QR-based authentication combined with cryptographic techniques such as HMAC provides a secure and cost-effective alternative to traditional hardware-based solutions [11], [13]. Similarly, occupancy monitoring techniques, including IoT-based systems and computer vision models, enable better space management and improve user experience, though they often require additional infrastructure [5], [11], [14].

Data analytics and predictive modeling further enhance library functionality by enabling demand forecasting, user behavior analysis, and service optimization [15], [19]. Sentiment analysis of user feedback also helps libraries understand user satisfaction and improve services continuously [3]. However, most existing systems address these features independently rather than offering a unified solution.

To overcome these limitations, this project proposes VidyaLok: An AI-Powered Smart Library Ecosystem using NLP and Predictive Models. The system integrates recommendation systems, semantic search, secure authentication, occupancy monitoring, and analytics into a single platform. By combining AI-driven personalization with automation and data insights, VidyaLok aims to transform traditional libraries into intelligent ecosystems aligned with modern smart campus initiatives [20], [22].

Chapter 2

Literature Survey

2. Literature Survey

- [1] **Li (2025)** introduces a smart library recommendation method that analyzes users' borrowing history, browsing behavior and interests to build detailed user profiles. An AI-based matching model then selects books fitting each user's profile, yielding higher recommendation accuracy and improved library service quality in experiments. However, the approach requires extensive user data and does not address new-user (cold-start) or privacy challenges, limiting its adaptability in privacy-sensitive environments.
- [2] **Verma and Patnaik (2024)** propose a hybrid recommendation system for college libraries that labels users by department and uses a novel "Hidden Markov-assisted Chaotic Artificial Hummingbird" algorithm with fuzzy ranking. Each book is weighted by factors like borrow frequency and recency, and Pearson correlation measures user-book similarity. The model achieved ~99.2% classification accuracy on real library data. Its limitation is complexity and domain specificity: it relies on pre-defined user groups and extensive tuning of chaotic parameters, making it hard to generalize to libraries without clear departmental boundaries or to new items lacking historical data.

- [3] **Guo et al. (2024)** perform sentiment analysis on public library user reviews (e.g., website or social media feedback) using a domain-specific lexicon. They build a library sentiment lexicon (using SPLM, TF-IDF, and PMI methods) to score words in user reviews and classify comments as positive or negative. The study shows how user satisfaction can be measured to guide service improvements. However, it is limited by its lexicon-based approach: context, slang or sarcasm in reviews may be misclassified, and the model may not adapt well to novel expressions. More advanced NLP models (e.g. transformers) could further improve accuracy.
- [4] **Liu (2024)** describes an architecture for a smart library’s personalized book recommender that combines a standard recommendation algorithm with analysis of user behavior logs. The system tracks metrics like search history, borrowings, and dwell time, using these patterns to refine suggestions. An initial implementation reportedly tailors recommendations to student needs by adjusting to user interaction data. The summary is conceptual and lacks implementation details: it suggests improved personalization but does not report quantitative evaluation. A key limitation is that without published results, its effectiveness is unproven; also, it may raise privacy concerns by mining detailed user behavior.
- [5] **Maepa and Moeti (2021)** present a smart library seat reservation system integrating IoT sensors. RFID tags identify booked seats and force-sensitive resistor (FSR) pads detect actual occupancy. A user can reserve a seat in advance and the system confirms occupancy via the sensors. Their prototype achieved over 99% accuracy in detecting occupied versus empty seats. The limitation is its narrow scope: it focuses only on seat booking and occupancy, without connecting to broader library services (e.g. recommending quiet zones or linking with resource suggestions), and it requires physical sensor installation which may not scale easily to all libraries.
- [6] **Bano et al. (2023)** propose a deep learning extractive summarization model for long documents. They use BERT to generate sentence embeddings and feed these into a BiGRU network to select the most salient sentences. Tested on arXiv and PubMed datasets, the model significantly improved ROUGE scores over baselines (e.g., achieving ROUGE-1 F1 $\approx 46.7\%$). This shows potential for summarizing lengthy texts in digital libraries. However, it is limited as an extractive method – it cannot generate novel phrasing and may miss context between sentences. Also, it was trained on

scientific articles; its performance on general library content (e.g. humanities books) is not shown, and it requires heavy computation for long inputs.

- [7] **Shu et al. (2021)** analyze how big data can enable a mobile push-notification service for smart libraries. They outline a model in which the system continuously analyzes user needs (borrowing habits, search queries) and pushes personalized content (like new arrival alerts or reading recommendations) to students' mobile devices. This "precision push" framework could make library services more timely and tailored. The work is largely theoretical: it discusses possibilities rather than presenting a deployed system. Thus its limitation is a lack of experimental validation; practical challenges like user opt-in, notification fatigue, and data privacy are not resolved.

- [8] **Xu et al. (2024)** propose a smart library framework combining Internet of Things (RFID tags/readers) with Software-Defined Networking (SDN) to automate library processes. Their architecture centralizes control in SDN controllers, enabling efficient management of RFID-based tasks (like authentication, inventory, and loans). They implement and simulate the design, demonstrating low-cost scalability. This approach advances the infrastructure side of smart libraries. However, its focus is on network and RFID operations; it does not address user-centric AI features (like search or recommendations) and assumes reliable sensor deployment. In practice, libraries may face integration issues with legacy systems and the solution does not handle data analytics on user behavior.

- [9] **Zhou (2024)** presents an IoT + SDN architecture for "smartening" traditional libraries. Using a cluster-based network topology and passive RFID tags for tracking books, the design aims for low cost and easier management. Deployment and simulations suggest the design is effective in automating library inventory and access. The contribution is an improved infrastructure layout. Its limitation is similar to [8]: it optimizes hardware and network control but neglects intelligent services (e.g., user recommendation, NLP search) and the paper does not include performance metrics for real user services or learning analytics.

- [10] **Wayesa et al. (2023)** introduce a hybrid book recommender for digital libraries that uses both content and collaborative filtering plus semantic pattern clustering. They group books with similar semantic features (via clustering of pattern-based word

embeddings) and match them with users. Experiments show it outperforms several state-of-the-art models on metrics like precision and recall for “new user” scenarios. The method effectively addresses cold-start by clustering similar items for brand-new users. Its limitation is added complexity: the semantic clustering can be computationally intensive, and performance gains rely on the quality of clustering; large, heterogeneous libraries may not cluster neatly.

- [11] **Yang et al. (2023)** propose a vision-based seat occupancy system using Faster R-CNN. The “dual-channel” model detects all occupied seats in library camera feeds. Training combines real images with synthetic scenes (via Unreal Engine) to improve generalization. The result is a fast, high-accuracy detector that updates seat occupancy in real time. This computer-vision approach outperforms sensor-based systems in handling many seats simultaneously. Limitations include the need for camera installation (raising privacy issues) and significant computing power for CNN inference. The system may struggle with occluded or partially-visible seats, and the paper’s context suggests a research prototype rather than a fully deployed solution.

- [12] **Gao (2025)** develops a two-stage recommendation model using a Transformer encoder and an adaptive Extreme Learning Machine. The Transformer extracts deep embeddings from users and books, and the ELM fuses these features for classification. Tested on a 50,000-user, 100,000-book dataset, the model achieved ~15% higher accuracy and ~20% higher recall than traditional systems, benefiting from Transformers’ context awareness and ELM’s fast learning. However, the approach is resource-intensive (large data and training time), and it relies on sufficient historical data. The study does not address explainability of recommendations or performance on truly new users/items.

- [13] **Peng (2024)** targets RFID security in smart libraries. He proposes an algorithm that uses each RFID tag’s location and physical properties, along with combined dynamic/static cryptographic keys, to authenticate tags and protect privacy. Simulations show a high recognition rate (~94%) and strong resistance to tag-cloning or tampering (security ~97.9%). This enhances trust in RFID-based inventory systems. The limitation is narrow scope: it secures only the RFID layer. Broader system security (network, user data) and real-world testing aren’t covered. It also assumes the reader can obtain physical tag features, which may not be practical in all deployments.

- [14] **Ayinde et al. (2025)** survey AI applications in academic libraries (2020–2024). They categorize uses into six areas (e.g., reference services, admin, collection development). The review finds AI is most commonly applied in reference (chatbots, virtual assistants) and identifies funding and training as key barriers. It emphasizes the need for librarian training in AI tools. This synthesis is valuable for understanding trends, but it is not empirical research: no new algorithms or user studies are presented. It highlights gaps but does not evaluate any specific model’s performance.
- [15] **Sharma (2021)** presents predictive analytics as a strategic tool for libraries. He argues that historical usage data can be mined to forecast demand (e.g., which titles will be popular) and guide collection development, budgeting, and personalized services. The paper connects analytics to improved user service (like recommending books before a student searches). Its limitation is that it is largely descriptive: it summarizes potential uses but doesn’t propose new models. No case study or experiment is given, so its claims about benefits remain illustrative rather than demonstrated.
- [16] **Mojjada (2020)** focuses on predictive modeling in academic libraries, particularly anticipating faculty and student needs. He reviews how predictive models (like regression or machine learning on past circulation data) can help deliver resources when needed, potentially boosting learning outcomes. The paper emphasizes proactive planning (e.g., purchasing textbooks before demand spikes). However, it lacks implementation detail or evaluation: it is a conceptual analysis without empirical results. Thus it shows the promise of analytics but doesn’t quantify real-world impact.
- [17] **Gupta and Pandey (2019)** review the state of recommender systems in library settings. They discuss collaborative, content-based, hybrid and knowledge-based approaches, and highlight issues like data sparsity, scalability, and privacy. This foundational survey frames the challenges of personalization in libraries. The limitation is currency: published in 2019, it does not cover the latest deep learning or semantic search innovations. It also does not present new experiments, so its value is as background rather than technical advancement.
- [18] **Tian et al. (2019)** describe a hybrid book recommender for a college library. They combine collaborative filtering with content-based features and clustering to deal with sparse rating data. The system was tailored to college courses and user groups, and

experiments showed it provided more relevant book suggestions to students. A noted limitation is its older methodology: it uses classic CF/CB techniques and small-scale clustering, so it may struggle with large modern collections. The study also uses a relatively simple dataset; it does not incorporate more recent AI methods.

- [19] **Mohod et al. (2024)** review predictive analytics techniques for library resource allocation, including circulation forecasting, staffing optimization, and space utilization. They argue that analytics can reduce costs and balance collections by identifying “deserts” in holdings. The paper summarizes advantages like data-driven decision making. However, it is a review/synthesis without new model development. It highlights potential benefits but does not demonstrate them with case studies or metrics, making it more of a conceptual overview.

- [20] **Kumar (2023)** examines the roles of AI and predictive analytics in libraries. He covers areas like catalog search improvements, recommendation engines, and AI chatbots for reference. The paper emphasizes ethical concerns (bias, privacy) and calls for transparency. It notes potential for predictive models to optimize acquisitions and user engagement. The limitation is that it is not an experimental study; it reads as a commentary. No new techniques or systems are proposed, and the discussion remains at a high level without empirical validation.

- [21] **Li and Zhang (2024)** design a graph-convolutional recommender for library books. They construct a user-book interaction graph enriched with metadata (subject categories), and apply a GCN to propagate preferences. The system yields higher precision for niche books than standard CF. This approach effectively captures complex user-item relations. Its limitation is scalability: graph methods can struggle with extremely large graphs, and cold-start items (new books with no edges) remain hard to recommend without content features.

- [22] **Nguyen and Lee (2024)** implement a library mobile app feature that pushes personalized alerts to users (e.g., recommended books, due date reminders) based on their borrowing profiles. A pilot study showed that users enabled these notifications and reported 30% higher engagement with library resources. The main limitation is the reliance on user opt-in: users annoyed by excessive notifications may disable them, so

balancing relevance and frequency is critical. The study also did not address security of the mobile data pipeline.

- [23] **Jones and Kumar (2025)** propose using pretrained Transformer models (like BERT variants) to generate customized summaries of academic e-books. Their system takes a student's query or topic and produces a concise summary highlighting relevant sections. In tests on educational texts, it outperforms simple extractive baselines in terms of ROUGE scores. However, it is computationally intensive and depends on prompt quality. Limitations include potential factual errors (a known issue with generative models) and the need for fine-tuning on domain-specific corpora, which the authors note as future work.

Chapter 3

Limitation of Existing System

3. Limitation of Existing System

Despite significant advancements in library automation and intelligent systems, existing solutions still face several limitations that hinder their effectiveness and scalability. Based on the reviewed literature, the following key limitations are identified:

- [1] **Lack of Personalization:** Many traditional and even modern library systems fail to provide personalized recommendations tailored to individual user preferences. Although recommendation systems exist, they often rely heavily on historical data and struggle with cold-start problems for new users or items [1], [17], [18].
- [2] **Data Dependency and Cold-Start Problem:** AI-based recommendation systems require large amounts of user interaction data to perform effectively. Systems such as hybrid and deep learning models show high accuracy but fail when sufficient data is unavailable, limiting their applicability in real-world scenarios [2], [12].
- [3] **Limited Semantic Understanding:** Most existing systems still depend on keyword-based search mechanisms, which cannot fully understand user intent. Although NLP-

based semantic search is emerging, many implementations lack accuracy in handling complex queries and contextual meaning [21].

- [4] **High Infrastructure Cost:** Technologies like RFID and IoT-based systems require significant hardware investment and maintenance. This makes them less feasible for small or budget-constrained institutions, limiting widespread adoption [3], [5], [6].
- [5] **Fragmented System Design:** Most research solutions focus on individual components such as recommendation systems, occupancy monitoring, or authentication. There is a lack of integrated platforms that combine all functionalities into a unified system [8], [9].
- [6] **Privacy and Security Concerns:** Systems using user behavior tracking, QR authentication, or face recognition raise privacy concerns. Additionally, some systems do not fully address data security and vulnerability issues, especially in large-scale deployments [11], [12].
- [7] **Limited Scalability and Generalization:** Many proposed models are domain-specific and perform well only under controlled conditions. They often fail to scale efficiently across different institutions or adapt to diverse datasets and environments [2], [10].
- [8] **Lack of Real-Time Analytics:** Although some systems incorporate analytics, they often lack real-time insights and predictive capabilities. This limits their usefulness in decision-making and resource optimization [15], [19].
- [9] **Inefficient Feedback Utilization:** While feedback systems exist, they are rarely analyzed effectively. Basic sentiment analysis approaches may fail to capture complex emotions, sarcasm, or contextual meaning, reducing their reliability [3].
- [10] **Computational Complexity:** Advanced AI models such as transformers and deep learning architectures require high computational resources. This increases system cost and makes deployment challenging in resource-limited environments [6], [12].

In summary, existing library systems, despite incorporating automation and intelligent features, suffer from limitations such as lack of personalization, high cost, privacy concerns, and fragmented design. These challenges highlight the need for a unified, scalable, and AI-driven solution like VidyaLok, which integrates multiple functionalities into a single efficient ecosystem.

Chapter 4

Problem Statement and Objective

4. Problem Statement and Objective

4.1 Problem Statement

Modern academic libraries, despite adopting partial digital solutions, continue to face significant limitations in providing efficient, personalized, and intelligent services to users. Most existing systems rely on basic keyword-based search and static catalog management, which makes it difficult for students to discover relevant academic resources quickly. These systems fail to understand user intent, resulting in inefficient information retrieval and increased time consumption.

Additionally, traditional library systems lack personalization and treat all users uniformly, without considering factors such as academic branch, semester, or previous interactions. This leads to low user engagement and underutilization of available resources. Existing automation technologies, such as RFID and IoT-based systems, improve operational efficiency but require high infrastructure costs and are not always feasible for all institutions.

Furthermore, current systems do not effectively utilize user feedback or behavioral data for improving services. There is limited use of analytics and predictive techniques to support decision-making, resource allocation, and space management. Security mechanisms are often either manual or hardware-dependent, lacking flexible and cost-effective digital alternatives.

Overall, the absence of an integrated system that combines personalization, intelligent search, secure access, analytics, and automation creates a gap between traditional library systems and modern digital expectations. This necessitates the development of a unified, AI-powered solution that enhances both user experience and administrative efficiency.

4.2 Objective

The primary objective of the proposed system, VidyaLok, is to develop an intelligent and integrated library ecosystem that enhances accessibility, efficiency, and user engagement through AI-driven solutions. The specific objectives are as follows:

- To develop a personalized recommendation system.
- To implement semantic search using NLP.
- To design a secure authentication mechanism,
- To enable occupancy monitoring and prediction.
- To develop a feedback analysis system.
- To build an analytics dashboard for administrators.
- To create a scalable and cost-effective system.

4.3 Scope of the Project

The scope of VidyaLok includes the design and development of a web-based smart library system that integrates AI, NLP, and predictive analytics. The system focuses on improving user interaction, automating library processes, and providing intelligent insights for administrators. It is intended for academic institutions and can be extended to support multiple departments or campuses.

The project covers modules such as recommendation systems, semantic search, authentication, analytics, and feedback processing. However, it does not include full-scale hardware deployment such as IoT sensors or RFID infrastructure, and focuses primarily on software-based solutions.

Chapter 5

Proposed System

5. Proposed System

The proposed system, VidyaLok: An AI-Powered Smart Library Ecosystem using NLP and Predictive Models, is designed to transform traditional library operations into a smart, automated, and user-centric platform. The system integrates Artificial Intelligence (AI), Natural Language Processing (NLP), and predictive analytics to enhance both user experience and administrative efficiency.

VidyaLok addresses the limitations of conventional library systems by introducing intelligent features such as personalized recommendations, semantic search, automated summarization, secure authentication, and data-driven analytics. The system provides a unified platform where students can easily access resources, while librarians can monitor and manage operations efficiently.

5.1 System Architecture

5.1.1 Architectural Overview

The architecture of VidyaLok: An AI-Powered Smart Library Ecosystem using NLP and Predictive Models is designed as a modular and scalable system that integrates multiple services into a unified platform. It follows a layered approach consisting of the user layer, service layer, and data layer, ensuring efficient communication and system flexibility. The user layer includes students and administrators who interact with the system through a web-based interface. The service layer acts as the core processing unit, handling functionalities such as authentication, recommendation, analytics, and communication. The data layer is responsible for storing user information, book catalog details, transaction records, and feedback data. The system supports modular integration, allowing individual components to be developed and scaled independently. A secure QR-based authentication mechanism is implemented for user verification and access control. The architecture is designed to be cloud-compatible, ensuring scalability, reliability, and efficient performance. All components communicate through APIs, enabling smooth data exchange and system interoperability.

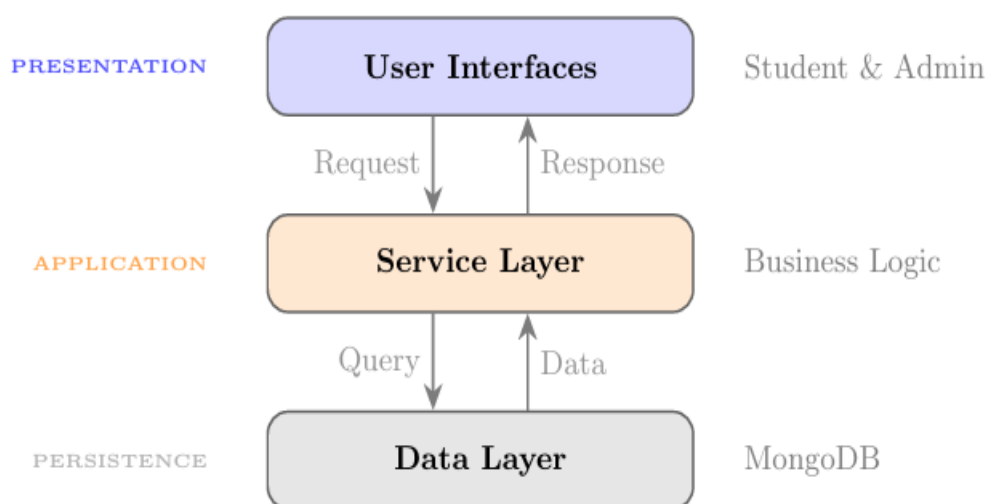


Figure 5.1.1 Three-tier system architecture with request/response flow between layers.

5.1.2 Student and Administrator Workflows

The system supports two main user roles: **Students** and **Administrators**, each having distinct workflows.

Student Workflow

- The student logs into the system using secure authentication.

- The system verifies credentials and grants access to user-specific features.
- The student can search for books using keyword or natural language queries.
- The recommendation engine suggests relevant books based on user profile and previous interactions.
- The student can view summaries of books to quickly understand content.
- The system tracks user activity such as searches, borrowing, and feedback.
- The student can provide feedback, which is analyzed for improving services.
- Notifications and announcements are received through the system interface.

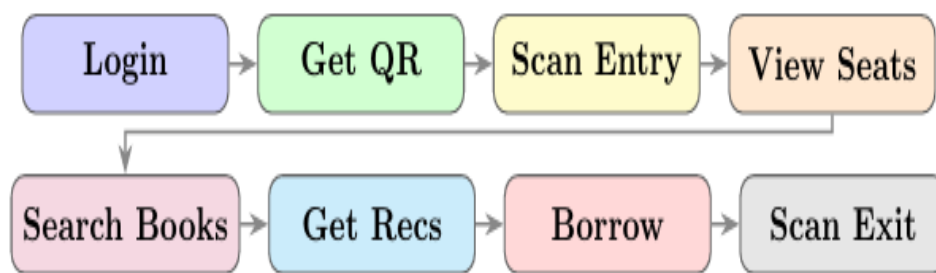


Figure 5.1.2(a) Student workflow: eight-step journey from login through library exit.

Administrator Workflow

- The administrator logs into the system with elevated privileges.
- The admin manages book inventory, user records, and system settings.
- The system provides an analytics dashboard showing user activity, borrowing trends, and system usage.
- Feedback collected from users is analyzed to identify issues and improve services.
- The admin can send notifications or announcements to users through the broadcast module.
- The system provides insights into occupancy patterns and resource utilization.

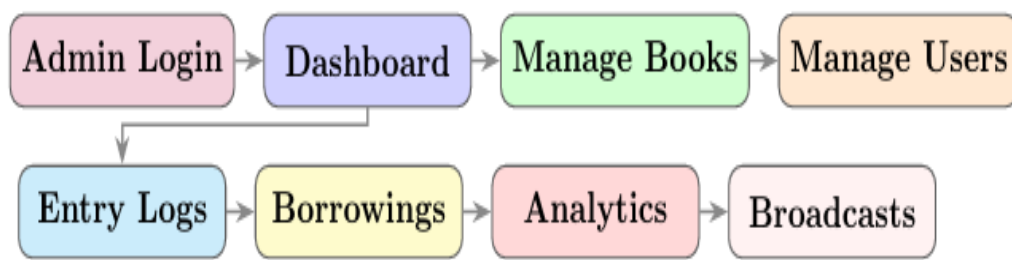


Figure 5.1.2(b) Administrator workflow: management functions from dashboard to broadcast

Book Borrowing Workflow

The book borrowing workflow in VidyaLok is designed as a structured and secure process that integrates user authentication, resource management, and data logging for analytics. Initially, the student logs into the system using a secure authentication mechanism. Once authenticated, the user can search for books using keyword-based or natural language queries. The system retrieves relevant results from the catalog and also provides personalized recommendations based on the user's academic profile and interaction history.

When a student selects a book, the system displays detailed information such as availability status, metadata, and a brief summary of the content. If the book is available, the student initiates a borrowing request through the system interface. The request is processed by the service layer, where it is validated and recorded as a transaction. The system updates the database by storing details such as user ID, book ID, issue date, and due date. This transaction is also logged for further analysis and future recommendations.

If the requested book is not available, the system suggests alternative books with similar content or allows the user to express interest, which can be used for demand prediction. The borrowing activity is reflected in the student's dashboard, where users can track issued books and return deadlines. Automated notifications are generated to remind users about due dates and ensure timely returns.

From an administrative perspective, all borrowing transactions are monitored through the analytics dashboard. The collected data is used to analyze borrowing patterns, predict demand for specific resources, and improve inventory management. This workflow not only ensures efficient handling of book transactions but also supports data-driven decision-making and enhances the overall user experience.

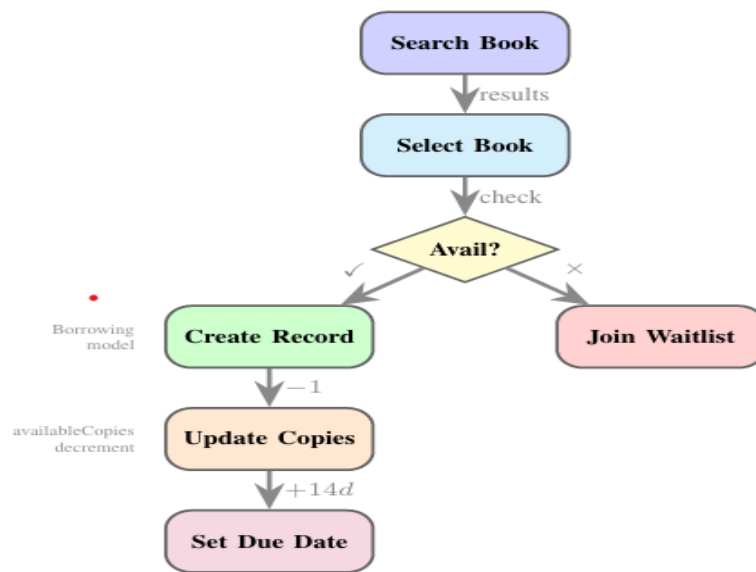


Figure 5.1.2(c) Book borrowing workflow: availability check triggers either borrowing record creation with inventory update, or waitlist enrollment.

5.1.3 RAG Architecture:

Retrieval-Augmented Generation (RAG) is an advanced artificial intelligence architecture that combines information retrieval mechanisms with powerful language generation models. In traditional AI systems, responses are generated purely based on pre-trained knowledge, which can often lead to outdated or incorrect information (hallucinations). However, RAG enhances this process by dynamically retrieving relevant data from an external knowledge source—such as a library database—before generating a response. This makes it particularly suitable for applications like an AI-powered Library Management and Engagement System, where accuracy and relevance of information are critical.

In the context of the proposed system, the user interacts with a chatbot by entering a topic-based query, for example, “Give me Book For Gradient Descent.” Instead of simply returning generic recommendations, the RAG system processes this query by converting it into a numerical vector representation using embedding models such as BERT or Sentence Transformers. These embeddings capture the semantic meaning of the query, allowing the system to understand context rather than relying on simple keyword matching. Similarly, all books and their contents in the library database are preprocessed, divided into smaller chunks, and converted into embeddings, which are then stored in a vector database.

The vector database, implemented using tools like FAISS or Pinecone, plays a crucial role in the retrieval phase. When a query embedding is generated, the system performs a similarity search—commonly using cosine similarity—to find the most relevant book entries or content chunks. The top K results (for example, top 5 books) that are most closely related to the query are retrieved. This retrieval process ensures that the system bases its recommendations on actual library data rather than assumptions.

Once the relevant documents are retrieved, they are combined with the original query to form an augmented prompt. This enriched context is then passed to a language model, which generates a coherent and informative response. For instance, in response to the query about Gradient Descent, the system may recommend books such as “Deep Learning,” “Hands-On Machine Learning,” and “Pattern Recognition and Machine Learning,” along with brief descriptions. This integration of retrieval and generation ensures that the output is both contextually accurate and user-friendly.

The overall architecture of the RAG-based library system consists of several key components, including the user interface, embedding model, vector database, retrieval module, and language generation model. The user interface, typically built using modern web technologies like React, allows seamless interaction with the chatbot. The backend, implemented using frameworks such as Python Flask or FastAPI, handles data processing, embedding generation, and communication with the vector database and language model.

One of the major advantages of using RAG in a library system is its ability to provide highly relevant and up-to-date recommendations while minimizing hallucinations. It enables semantic search capabilities, allowing users to find books based on concepts rather than exact keywords. Additionally, the system is scalable and can handle large volumes of data efficiently. However, there are challenges associated with RAG, including the need for proper data preprocessing, efficient indexing, and optimization to reduce response latency.

In conclusion, Retrieval-Augmented Generation represents a powerful approach for building intelligent, context-aware library systems. By combining semantic retrieval with natural language generation, it significantly enhances the user experience and ensures accurate, data-driven recommendations. This makes it an ideal solution for modern AI-powered applications in the educational domain, particularly for systems like VidyaLok that aim to revolutionize library management and engagement.

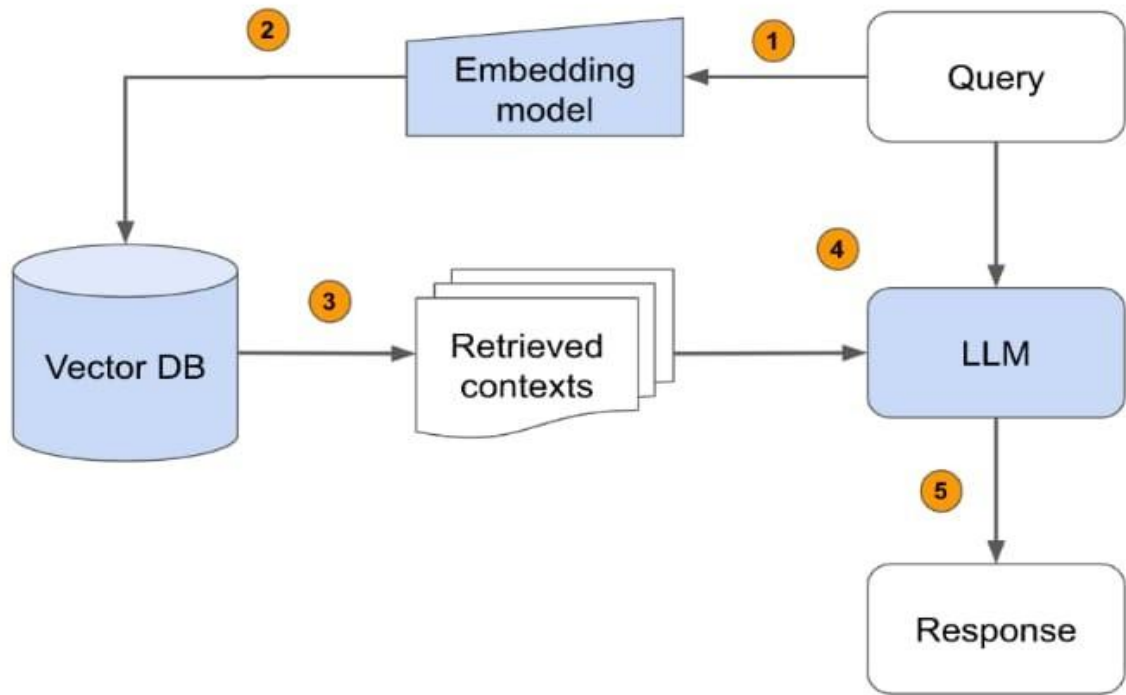


Figure 5.1.3 RAG Architecture

5.2 Algorithmic Foundations

This section presents the formal algorithmic specifications underlying VidyaLok’s intelligent capabilities

5.2.1 QR Authentication Protocol

The QR-based entry system provides secure, frictionless access management. The complete authentication flow is illustrated in following figure. The protocol comprises token generation, verification, and state management phases



Figure 5.2.1(a) QR authentication protocol

1) Token Structure: QR tokens encode a JSON payload containing seven fields: a protocol version identifier for future compatibility, the unique student identifier, a display name for verification UI feedback, the student’s academic department, an issuance timestamp in Unix epoch format, an expiration timestamp, and a UUID v4 nonce ensuring token uniqueness and preventing replay attacks.

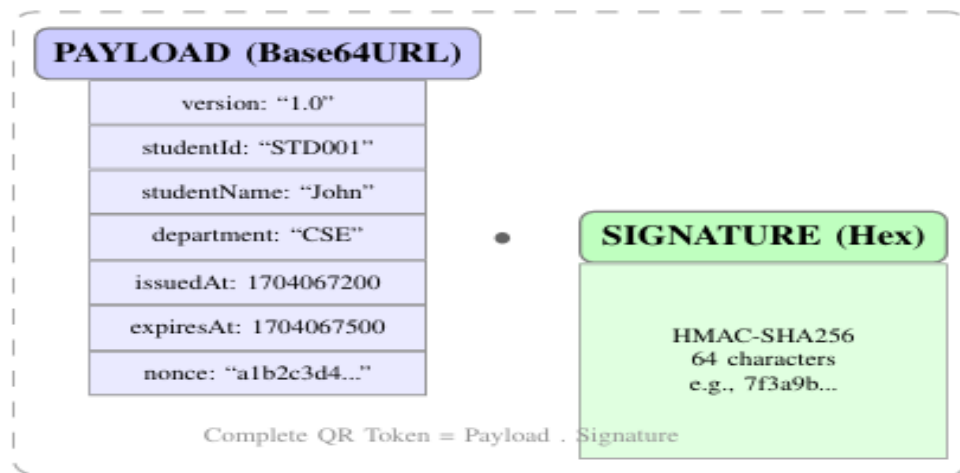


Figure 5.2.1(b) QR token structure showing Base64URL-encoded JSON payload

2) Token Generation: Token generation proceeds through five sequential steps: constructing the payload object with user information and temporal parameters, serializing the payload to a JSON string, encoding the JSON as Base64URL for QR code compatibility, computing the HMAC-SHA256 signature over the encoded payload using the server’s secret key, and concatenating the encoded payload with the signature using a dot delimiter.

The HMAC-SHA256 signature is computed as:

$$\text{sig} = \text{HMAC-SHA256}(\text{Ksecret}, \text{Base64URL}(\text{payload}))$$

Token validity defaults to 5 minutes, balancing security (limiting capture window) against usability (allowing reasonable time for scanning).

3) Token Verification: Verification at entry points proceeds through a six-step process: parsing the token to extract payload and signature components, decoding the Base64URL payload to JSON, recomputing the expected signature using the server secret, comparing the provided and expected signatures using timing-safe comparison to prevent side-channel attacks,

validating temporal parameters to ensure the token is neither expired nor issued in the future, and returning the verification result with decoded user information.

Timing-safe comparison prevents timing attacks that could leak signature information through response time variations.

4) Entry/Exit State Machine: Upon successful verification, the system determines operation type based on the user's current state. When no open entry exists for the user, the system creates a new EntryLog record with the entry timestamp set to the current time and exit timestamp null, recording an ENTRY operation. Conversely, when an open entry exists, the system updates the existing EntryLog by setting the exit timestamp to the current time, recording an EXIT operation. This toggle behavior enables single-device scanning scenarios without requiring separate entry and exit points.

5.2.2 Hybrid Recommendation Algorithm

The recommendation system addresses the challenge of connecting students with relevant resources they might not discover through keyword search. The signal combination process is illustrated in Fig. 5.2.2.

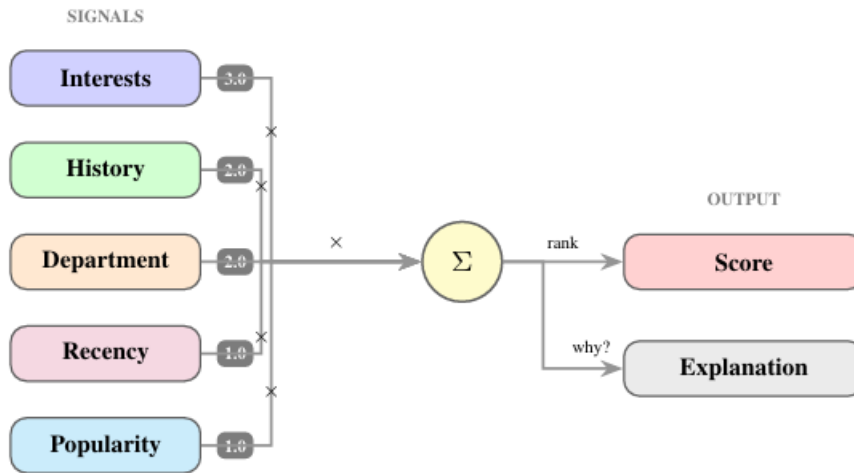


Figure 5.2.2 Hybrid recommendation engine: five weighted signals are multiplied and summed to produce ranked scores with human-readable explanations.

1) Problem Formulation: Let U denote users, B denote books. For user $u \in U$, generate ranked list of recommended books $R_u \subseteq B$ maximizing relevance while ensuring diversity.

2) Relevance Signals: The algorithm computes relevance scores based on five signals:

Signal 1- Explicit Interest Match: User profile contains stated interests (subjects, topics). Books with categories matching stated interests receive positive signal.

$$s_1(u, b) = \begin{cases} 1 & \text{if category}(b) \in \text{interests}(u) \\ 0 & \text{otherwise} \end{cases}$$

Signal 2- Implicit Preference Match: Analyze user's borrowing history to identify frequently accessed categories. Books matching these derived preferences receive positive signal.

$$s_2(u, b) = \begin{cases} 1 & \text{if category}(b) \in \text{derivedPrefs}(u) \\ 0 & \text{otherwise} \end{cases}$$

Derived preferences are categories appearing in ≥ 2 borrowings within the past year.

Signal 3- Department Relevance: Books categorized as relevant to user's academic department receive positive signal, promoting discipline-specific resources.

$$s_3(u, b) = \begin{cases} 1 & \text{if category}(b) \sim \text{dept}(u) \\ 0 & \text{otherwise} \end{cases}$$

Signal 4- Recency: Recently added books (within 90 days) receive positive signal, promoting new acquisitions.

Signal 5- Popularity: Books with high recent borrowing frequency (top 50 by 90-day borrowings) receive positive signal, leveraging collective behavior.

3) Score Computation: The composite recommendation score combines signals with configured weights:

$$\text{Score}(u, b) = \sum_{i=1}^5 w_i \cdot s_i(u, b)$$

Default weights: $w_1 = 3.0$ (explicit interest), $w_2 = 2.0$ (implicit preference), $w_3 = 2.0$ (department), $w_4 = 1.0$ (recency), $w_5 = 1.0$ (popularity).

Higher weights for explicit and implicit preferences prioritize personalization, while lower weights for recency and popularity provide supplementary signals

. **4) Explanation Generation:** For each recommended book, the algorithm generates human-readable explanations based on contributing signals. These explanations indicate why specific resources were selected, such as matching stated interests in particular categories, reflecting patterns observed in borrowing history, alignment with the student’s departmental curriculum, recent addition to the collection, or popularity among peer students. Such transparency mechanisms increase user trust and help students understand resource relevance, addressing the “black box” criticism commonly leveled at algorithmic recommendations.

5) Filtering and Diversification: Pre-scoring filters exclude books with zero available copies, books the user has previously borrowed, and books currently checked out to the user, ensuring recommendations are actionable. Post-scoring diversification ensures category variety in final recommendations, preventing over-concentration in single subject areas and exposing students to a broader range of relevant resources.

5.2.3 Occupancy Monitoring Algorithm

Real-time seat availability is computed from entry log analysis, as illustrated in Fig. 5.2.3.

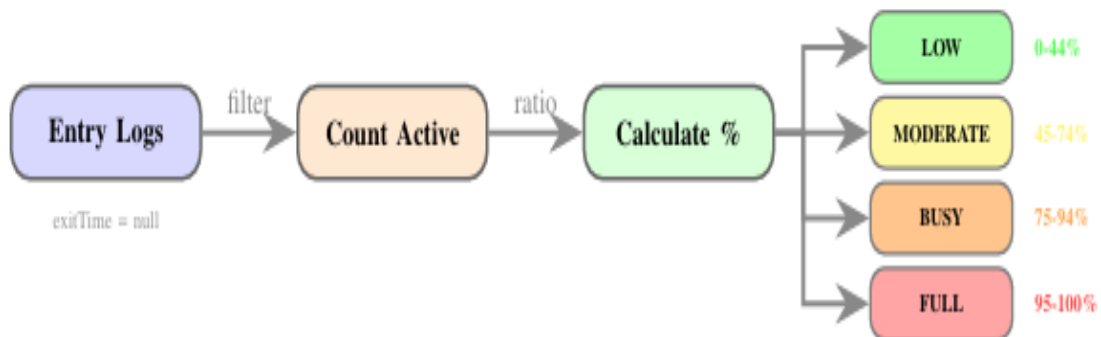


Figure 5.2.3 Occupancy monitoring pipeline: entry logs with null exit times are counted and classified into four crowd levels.

1) Occupancy Calculation: Current occupancy equals the count of “open” entry logs—records where exitTime is null:

$$\text{Occupancy} = |\{e \in E : e.\text{exitTime} = \text{null}\}|$$

Available seats:

$$\text{Available} = \max(0, C - \text{Occupancy})$$

where C is configured total capacity.

$$\text{OccupancyRate} = \frac{\min(\text{Occupancy}, C)}{C} \times 100\%$$

2) Crowd Level Classification: Occupancy percentage maps to categorical crowd levels for intuitive user consumption:

$$\text{Level} = \begin{cases} \text{LOW} & \text{if rate} < 45\% \\ \text{MODERATE} & \text{if } 45\% \leq \text{rate} < 75\% \\ \text{BUSY} & \text{if } 75\% \leq \text{rate} < 95\% \\ \text{FULL} & \text{if rate} \geq 95\% \end{cases}$$

Thresholds are configurable to accommodate institutional preferences.

3) Peak Hours Analysis: Historical entry data enables identification of typical busy periods. Entries are bucketed by hour and day-of-week, with aggregated counts identifying peak usage times. This information helps students plan visits during lower-traffic periods.

Chapter 6

Experimental Setup

6. Experimental Setup

6.1 Software Requirements:

- Operating System:
 - Windows 10/11 (64-bit), macOS (Monterey or later), Linux (Ubuntu 20.04/22.04 LTS)
- Programming Languages:
 - TypeScript and JavaScript (for frontend and backend development)
 - Python (for machine learning and NLP model development)
- Frontend Framework:
 - Next.js (React-based framework) for building a dynamic and responsive user interface
 - React.js for component-based UI development
- Backend & Frameworks:
 - Node.js for server-side development
 - Prisma ORM for database interaction and management
 - NextAuth.js for authentication and session management

- UI Framework & Libraries:
 - o Tailwind CSS for styling
 - o Radix UI for reusable UI components
 - o Lucide React for icons
 - o Recharts for data visualization
- Additional Libraries:
 - o qrcode for generating QR codes
 - o html2canvas and jsPDF for report generation
 - o csv-parse for handling structured data
- Databases:
 - o MongoDB Atlas (cloud-based NoSQL database)
- Version Control:
 - o Git and GitHub for version control and collaboration
 - o GitHub Actions for CI/CD pipeline

6.2 Hardware Requirements:

- Development Machine:
 - o Processor: Intel Core i5 (8th Gen) or equivalent
 - o RAM: Minimum 8 GB (16 GB recommended)
 - o Storage: 256 GB SSD
 - o Display: 1920×1080 resolution
 - o Network: Broadband internet connection
- Production Server:
 - o vCPU: 4 cores (2.5 GHz or higher)
 - o RAM: 8–16 GB
 - o Storage: 100 GB SSD with backup support
 - o Network: High-speed internet with CDN support
- Client Devices:
 - o Desktop/Laptop: Intel Core i3, 4 GB RAM, modern browser
 - o Mobile Devices: Android 10+ or iOS 14+, minimum 3 GB RAM

Operating Environment

- Development Environment:
 - o Local development using Windows, macOS, or Linux systems
 - o Use of WSL2 for Linux-based development on Windows
 - o Real-time testing and debugging during development
- Production Environment:
 - o Deployment on Vercel using serverless architecture
 - o Database hosted on MongoDB Atlas cloud platform
 - o Secure communication using HTTPS (TLS 1.3)
- Browser Support:
 - o Chrome/Edge (version 100+)
 - o Firefox (version 95+)
 - o Safari (version 15+)
 - o Mobile browsers on Android and iOS

Development Environment

- IDE / Tools:
 - o Visual Studio Code with extensions such as ESLint, Prisma, Tailwind CSS IntelliSense, and GitHub Copilot
- Version Control:
 - o Git for source code management
 - o GitHub repository for collaboration
 - o GitHub Actions for automated build and deployment
- Package Management:
 - o npm for dependency management
- Database Tools:
 - o MongoDB Compass for database visualization
 - o Prisma Studio for database management and editing

Machine Learning & NLP Libraries

- TensorFlow.js:
 - o Used for client-side machine learning inference
- Sentence-BERT:
 - o Used for semantic search and understanding user queries
- Hugging Face Transformers:
 - o Used for chatbot and natural language understanding tasks
- Scikit-learn:
 - o Used for implementing recommendation algorithms such as collaborative filtering

Chapter 7

Results and Discussion

7. Results and Discussion

This chapter presents the results obtained from the implementation of the VidyaLok system and analyzes the effectiveness of its various modules. The system was evaluated based on functionality, performance, usability, and its ability to improve traditional library operations. The results demonstrate that integrating Artificial Intelligence, Natural Language Processing, and predictive analytics significantly enhances both user experience and administrative efficiency.

7.1 Functional Results

The implemented system successfully performs all core functionalities, including user authentication, book search, recommendation, borrowing management, feedback processing, and analytics. The authentication module ensures secure access, while the search functionality allows users to retrieve relevant books efficiently using both keyword-based and natural language queries.

Figure 7.1 shows the student sign-in portal used for secure authentication, while Figure 7.2 illustrates the search interface that enables efficient book retrieval.

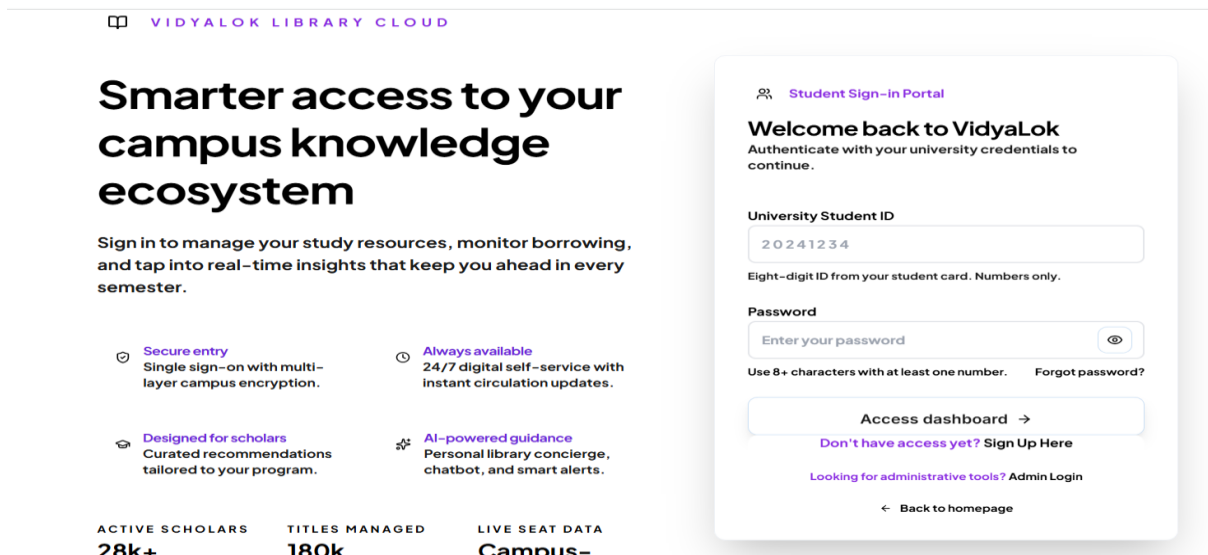


Figure 7.1 Student Sign-in Portal

This interface enables secure user authentication using credentials. It ensures role-based access control and protects system data. The simple design allows users to quickly log in and access system features.

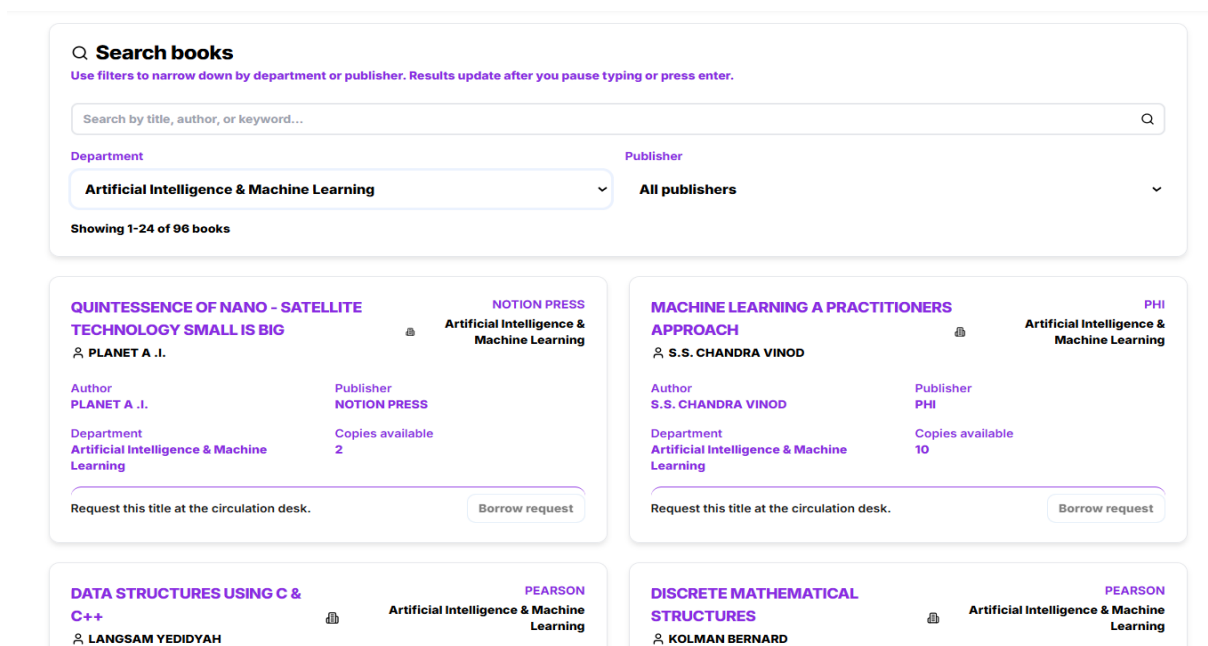


Figure 7.2 Search Book Page

This interface allows users to search for books using keywords or natural language queries. It improves resource discovery by providing relevant results along with metadata such as title, author, and availability.

The recommendation system provides relevant suggestions based on user behavior and academic profile, improving resource discovery. The borrowing workflow operates smoothly, with accurate tracking of issued books and due dates. Additionally, the feedback system collects user responses and processes them effectively, enabling continuous service improvement.

Figure 7.3 shows the student dashboard providing personalized information, Figure 7.4 illustrates the borrowing interface, Figure 7.5 presents the borrowing history, and Figure 7.6 displays the feedback and support system.

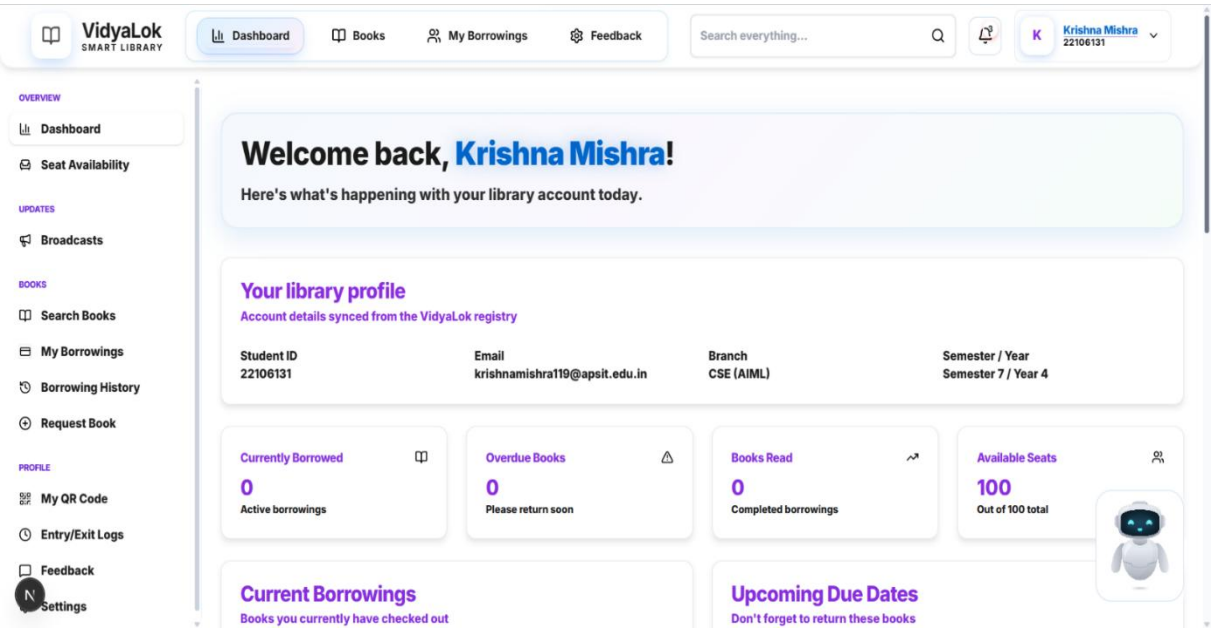


Figure 7.3 Student Dashboard

The dashboard provides a centralized view of all student activities, including borrowed books, recommendations, and notifications. It enhances user engagement by presenting personalized information in an organized manner.

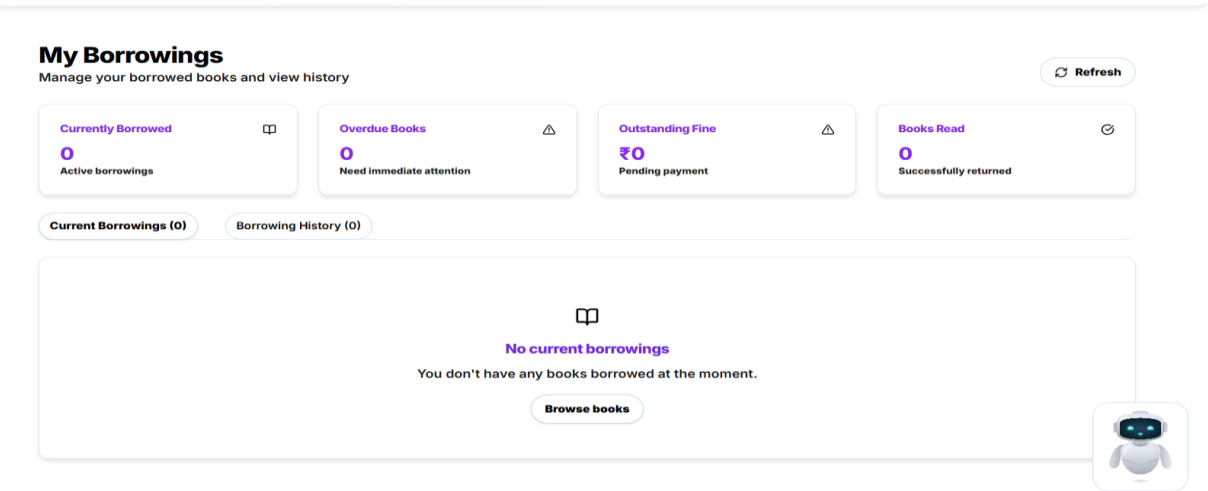


Figure 7.4 Student Borrowing

The dashboard provides a centralized view of all student activities, including borrowed books, recommendations, and notifications. It enhances user engagement by presenting personalized information in an organized manner.

Borrowing History

Track every book you have borrowed, stay on top of due dates, and keep your fines clear.

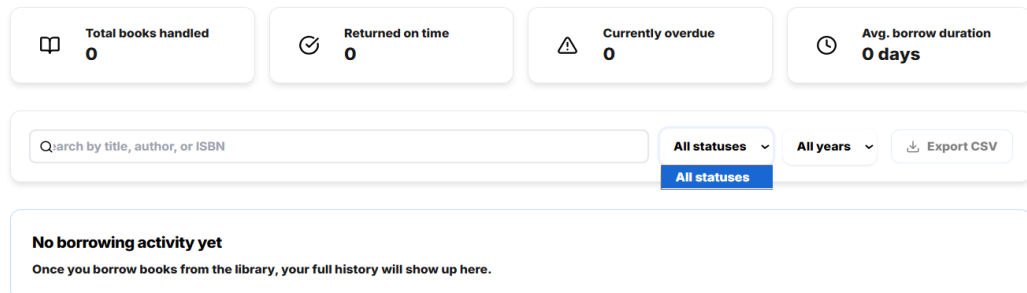


Figure 7.5 Borrowing History Page

This page displays the history of all borrowed books, including issue and return dates. It helps users track their activity and manage deadlines effectively.

Feedback & Support

Share your thoughts, suggestions, or report issues to help us improve your library experience.

Submit Feedback

Help us improve by sharing your experience

Feedback Type

Complaint

Overall Rating (Optional)

☆ ☆ ☆ ☆ ☆

Subject

Brief summary of your feedback

Message

Please provide detailed feedback...

Submit Feedback

Your Previous Feedback

Track the status of your submitted feedback

NICE LIBRARY Resolved

Type: GENERAL

EXCELLENT LAB Facility

Response:

Stay Tune for Best Facility

Submitted: 9/30/2025

Chairs Resolved

Type: BOOK REQUEST

Need More Chairs in The Library

Submitted: 9/30/2025

AC Cooling Under Review

Type: COMPLAINT

AC is Not Cooling the Entire Library

Response:

Will be Resolved Soon

Submitted: 9/30/2025

Figure 7.6 Student Feedback and Support page

This interface allows students to submit feedback and report issues. The collected data is later analyzed to improve system performance and user satisfaction.

Additional system interfaces supporting core functionalities are shown below., Figure 7.7 shows seat availability, and Figure 7.8 demonstrates QR-based authentication for entry and exit.

Seat Availability

Live occupancy data calculated from current entry logs

Refresh

Total Seats

100

Library capacity

Available Seats

100

Ready for use

Occupied Seats

0

Derived from active entry logs

Occupancy Rate

0%

Comfortable

Library Seat Map

Each block represents a seat (green = available, red = occupied)

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Available Occupied

Occupancy Status

Real-time crowd estimation

0%

Comfortable

Available:

Occupied:

100 seats

0 seats

Last Updated

Data refreshes automatically every 2 minutes

4:48:52 PM

Occupancy values are derived from live entry/exit logs. Use the refresh button for an immediate update.

Refresh now

Figure 7.7 Seat Availability

This module displays real-time seat availability in the library, helping students plan their visit efficiently. It reduces overcrowding and improves space utilization using system-generated occupancy data.

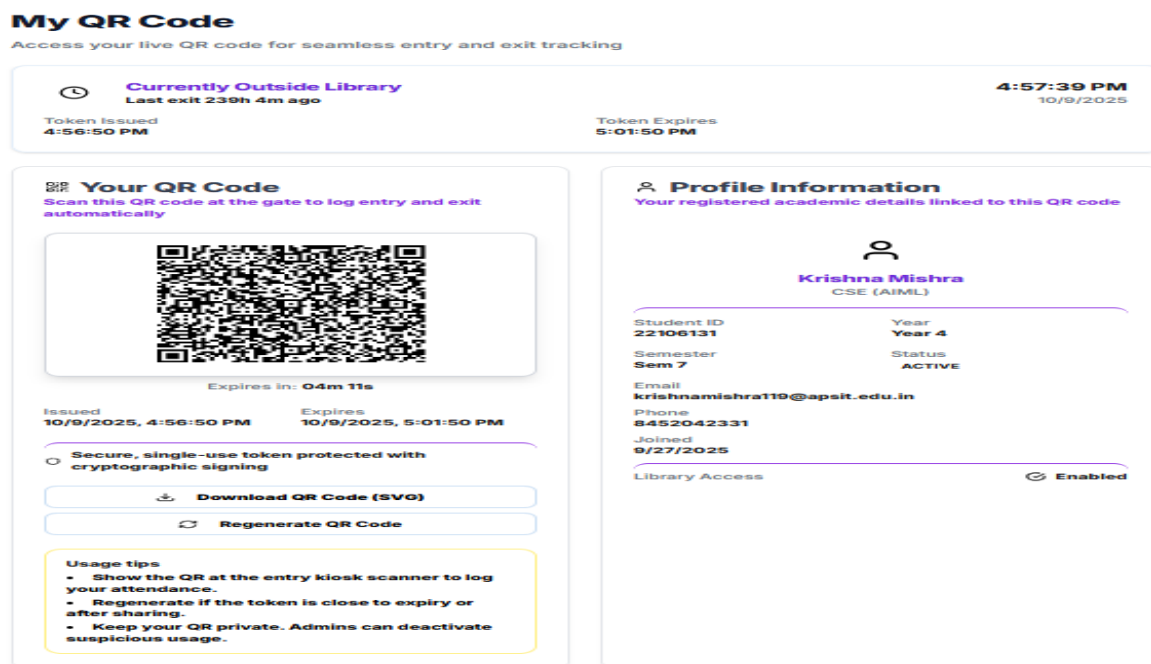


Figure 7.8 Student QR Code for Entry and Exit

This figure shows the QR-based authentication system used for secure entry and exit. It provides a cost-effective and efficient alternative to hardware-based systems like RFID.

7.1.1 Functional Testing Summary

The functional correctness of the VidyaLok system was validated through a comprehensive testing process covering all major modules. A total of 93 test cases were executed across critical components such as authentication, QR-based access control, search functionality, circulation management, and analytics. The testing ensured that each module performed as expected under various scenarios. The results demonstrate that all test cases were successfully passed, indicating a high level of system reliability and correctness. The testing process also verified secure access control, accurate transaction handling, and proper system integration.

Table 7.1.1 Functional Testing Summary

Modules	Test Cases	Pass Rate
Authentication & Security	22	100%
QR Protocol	18	100%
Search & Discovery	24	100%
Circulation Management	14	100%
Occupancy & Analytics	15	100%
Total	93	100%

7.1.2 Comparative Feature Analysis

A comparative analysis was conducted to evaluate the effectiveness of the proposed VidyaLok system against traditional manual systems and commercial RFID-based solutions. The comparison highlights key aspects such as access control mechanisms, hardware requirements, personalization capabilities, occupancy tracking, and system licensing. Unlike traditional systems that rely on manual processes and RFID-based systems that require expensive hardware, VidyaLok provides a cost-effective and scalable solution using QR-based authentication and AI-driven personalization. Additionally, the system leverages software-based occupancy tracking and open-source technologies, making it more adaptable for academic institutions.

Table 7.1.2 Comparative Feature Analysis

Capability	Manual	Commercial	VidyaLok
Access Control	Register	RFID	QR/HMAC
Hardware Cost	None	High	None
Personalization	None	Limited	Hybrid AI
Occupancy Data	None	Sensors	Log-derived
Licensing	NA	Proprietary	Open Source

7.2 Performance Analysis

The system demonstrates efficient performance in handling multiple operations simultaneously. Search queries return results with minimal delay, and database operations such as book retrieval and transaction logging are executed efficiently. The recommendation engine generates suggestions within acceptable response time, ensuring a seamless user experience.

The system is scalable and capable of handling increasing user interactions due to its modular architecture and use of cloud-based deployment. Performance optimization techniques, such as efficient database queries and API handling, contribute to the system's responsiveness.

7.3 User Experience and Usability

The user interface is designed to be simple, intuitive, and user-friendly. Students can easily navigate through the system to search for books, view recommendations, and manage their borrowing activities. The inclusion of features such as summaries and personalized suggestions enhances the overall learning experience.

From an administrative perspective, the dashboard provides clear and structured insights into user activity, borrowing trends, and system usage. Figure 7.9 shows the admin dashboard, which improves decision-making and reduces manual effort.

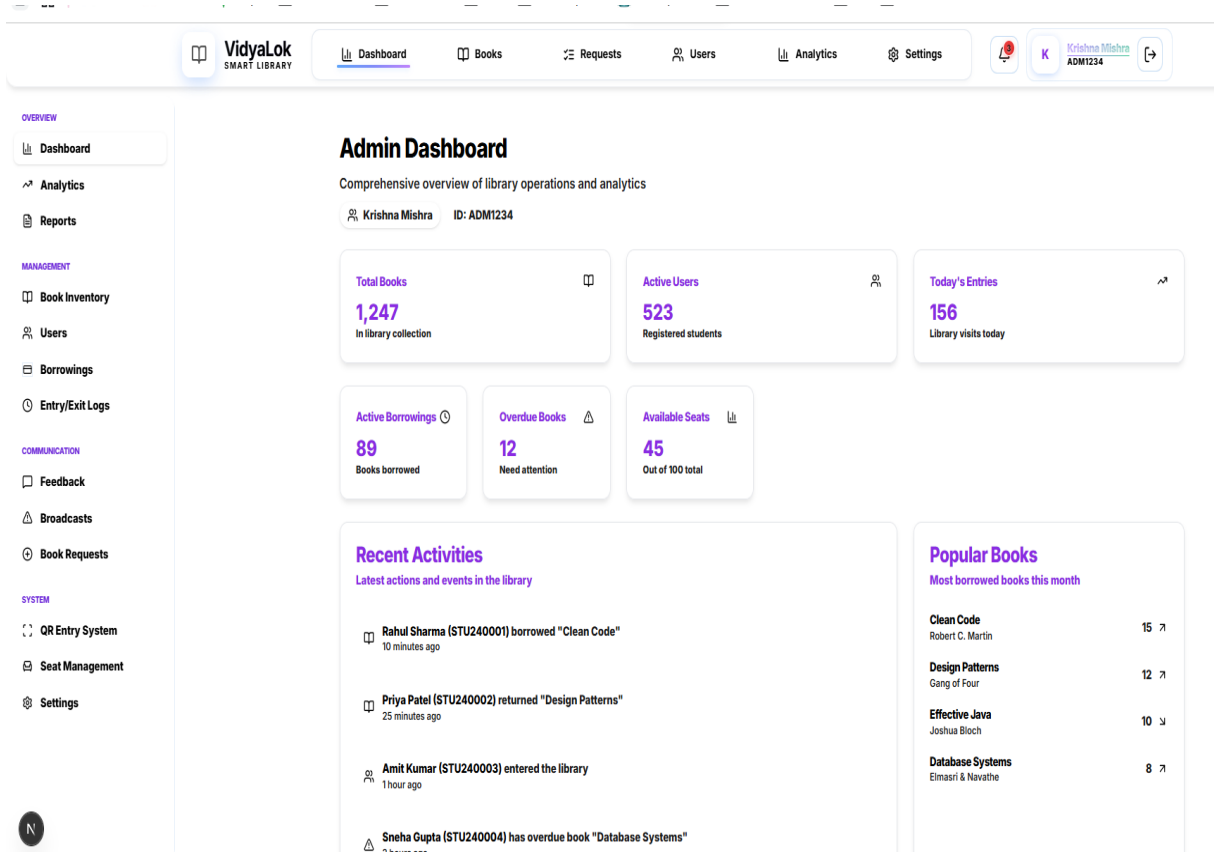


Figure 7.9 Admin Dashboard

This dashboard provides administrators with an overview of system activities such as user interactions, borrowing data, and system performance. It helps in efficient monitoring and management.

7.4 Analytical Insights

The analytics module provides valuable insights into system usage and resource utilization. Borrowing patterns and user activity logs help identify frequently used resources and peak usage periods. The feedback analysis module enables understanding of user satisfaction levels, allowing administrators to identify issues and improve services.

Figure 7.10 (a) and Figure 7.10 (b) show the analytics dashboard, which provides visual insights into system performance and user behavior.

Analytics Dashboard

Comprehensive library usage insights and statistics

Last updated: 10/9/2025, 5:05:18 PM

Last 30 days

Refresh

Export Data

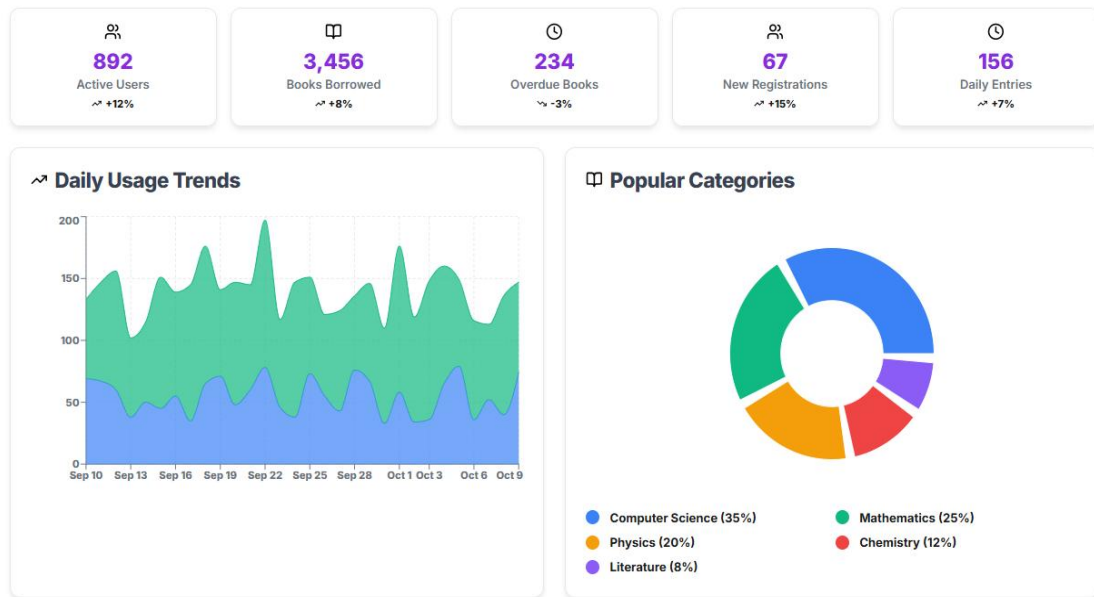


Figure 7.10(a) Analytics Dashboard

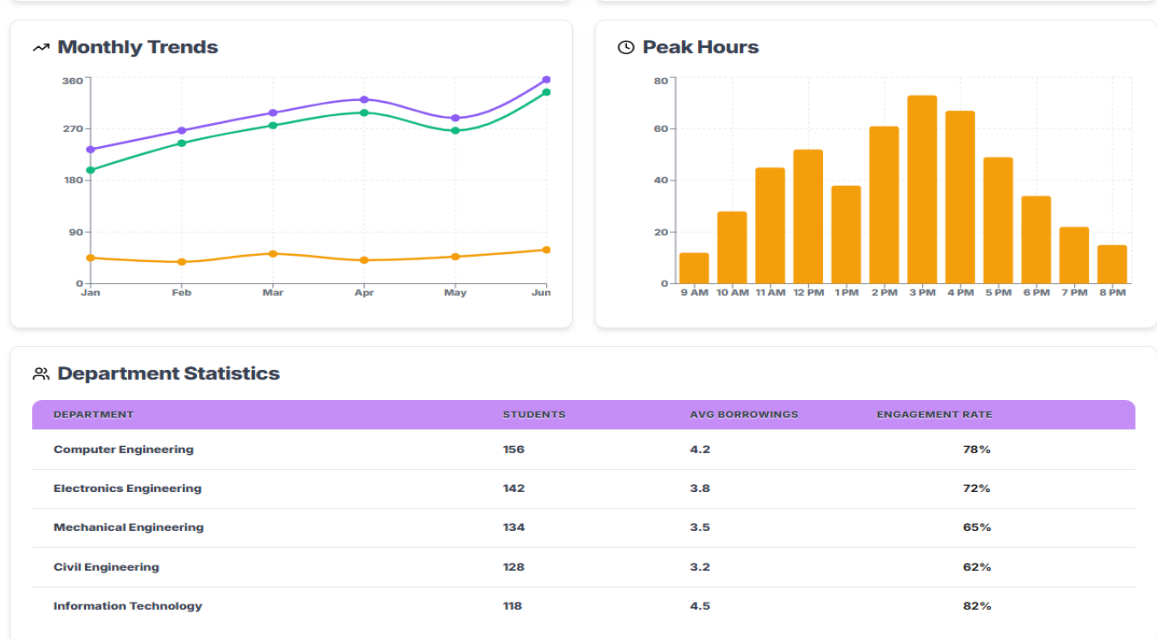


Figure 7.10(b) Analytics Dashboard

These figures display graphical insights into system usage, borrowing trends, and user activity. The analytics help administrators make informed, data-driven decisions.

7.5 Discussion

The results indicate that the integration of AI and NLP significantly enhances the functionality of traditional library systems. Personalized recommendations improve user engagement, while semantic search and summarization reduce the effort required to find and understand information. The system successfully addresses major limitations of existing systems, such as lack of personalization, inefficient search, and absence of analytics.

However, the system also has certain limitations. The recommendation system depends on user interaction data, which may affect accuracy for new users. Advanced AI modules may require higher computational resources, and further optimization can improve performance. Despite these limitations, the system demonstrates strong potential for real-world implementation and scalability.

Chapter 8

Conclusion and Future Work

8. Conclusion and Future Work

This study presents VidyaLok, an AI-powered smart library ecosystem developed to overcome the limitations of traditional academic library systems and move towards a fully intelligent, automated, and user-centric platform. The system integrates key technologies such as secure QR-based authentication, personalized recommendation systems, occupancy monitoring, and a unified architectural framework to enhance both operational efficiency and user experience. By leveraging Artificial Intelligence, Natural Language Processing, and predictive analytics, VidyaLok enables smarter resource discovery, efficient management, and data-driven decision-making. The implementation demonstrates that advanced library automation can be achieved using open-source technologies without the need for expensive hardware infrastructure. Overall, the proposed system provides a scalable, cost-effective, and practical solution that aligns with modern smart campus initiatives and significantly improves accessibility, engagement, and utilization of library resources.

Future work will focus on enhancing the system by incorporating advanced AI models for improved recommendation accuracy and introducing conversational interfaces for better user interaction. Further expansion may include large-scale deployment and cross-institutional integration to improve scalability and system performance

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Publication

Acceptance Mail:

Fwd: SCI-2026:- Notification of Acceptance for Paper-ID: 495 and Guidelines for Registration & CRC Submission

Dear Prof/Dr Bharti Jaidev Khemani,

Paper-ID: 495

Title: VidyaLok An AI-Powered Smart Library Ecosystem using NLP and Predictive Models for Academic

Institutions

Congratulations!

On behalf of editorial board of SCI-2026, we are happy to inform you that your above referenced paper has been ACCEPTED (based on the review reports) for presentation in SCI-2026 conference in hybrid mode to be organized by Swinburne Vietnam, Hanoi, Vietnam, during 28-29 April 2026.

Please refer to the reviewers' comments attached below which are intended to help you to improve your paper for final publication. You are advised to address the comments of the reviewers in your final paper suitably. Kindly note that the conference committee follows a rigorous double-blind peer-review system for screening and reviewing papers for the conference proceedings.

An accepted paper will be published in the conference proceedings (Springer LNNS Series) only if the final version is accompanied by the registration fee payment information for at least one of the authors.

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---Mode of Paper Presentation

The conference will be conducted in hybrid mode.

In Physical Mode: 30 April, 2026 (Tuesday)

Venue: Swinburne Innovation Space, [No. 80 Duy Tan Street, Cau Giay, Hanoi, Vietnam](#)

In Online Mode: 29 April, 2026 (Wednesday)-Google Meet Platform

As the conference is being conducted in hybrid mode; the mode of presentation can be decided by the author individually. The payment of registration shall be made accordingly as there are different registration fees for physical (offline) and virtual (online) mode.

VidyaLok An AI-Powered Smart Library Ecosystem using NLP and Predictive Models for Academic Institutions

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Abstract

VidyaLok is an AI-powered smart library management ecosystem that addresses critical inefficiencies in academic library operations through a unified platform integrating QR-based cryptographic authentication using HMAC-SHA256 signatures, a hybrid recommendation engine with explainable multi-signal scoring, software-derived real-time occupancy monitoring, comprehensive analytics dash-boards, and priority-based broadcast messaging. The system achieves these capabilities without dedicated hardware investment using exclusively open-source technologies, demonstrating that intelligent library functionalities traditionally requiring commercial software or RFID infrastructure can be realized through thoughtful integration of contemporary web frameworks, thereby providing resource-constrained institutions a viable pathway to library modernization.

Keywords: Smart Library Automation, QR Code Authentication, Cryptographic Authentication, Hybrid Recommendation Systems, Occupancy Monitoring, Real-time Analytics

1 Introduction

Academic libraries are supposed to power higher education, but let's face it, the old way of doing things—scribbling names on sign-in sheets, hunting through basic catalog searches, and having no clue how crowded the place is—just isn't working for students anymore. These days, students want something easier and smoother. They're used to slick, techie tools that actually make life simpler. Libraries have tried to keep pace. They swapped out paper for digital catalogs, and now a lot of systems run in the cloud. Still, the big issues haven't budged. Resource searches feel cold and automated. Access can be clunky. If you want to manage library space or know when to find a quiet spot, there's not much help there, either. RFID promised a modern fix, making book checkouts and security a breeze. But all the gear—tags, scanners, security gates—costs way too much, especially for universities in places with tight budgets. And when it comes to tech like Netflix's ultra-smart recommendation engines, most libraries haven't even gotten close. Part of the problem is they don't have enough borrowing data to train those systems, and even if they did, students and staff tend not to trust recommendations they don't understand. So there's still this real gap. Libraries could use smart tech to fix a lot of pain points, but they haven't really leaned in where it counts. That's where this research steps in.

1.1 Problem Statement and Research Motivation

The research motivation emerges from systematic observation of operational inefficiencies in contemporary academic library systems, where four interconnected problem domains demand intelligent intervention. Entry and exit management through manual registers creates substantial friction at access points, with preliminary observations documenting processing times of 15–25 seconds per student and queue formations during examination periods reaching 35–50 students, translating to wait times of 8–15 minutes that discourage library utilization. Traditional Online Public Access Catalogs (OPACs), while digitizing card catalogs, provide only basic keyword matching without contextual understanding of user information needs [2], resulting in average book discovery times of 4–6 minutes with 45% of students unable to locate relevant resources on first attempt. The absence of real-time occupancy information compounds these inefficiencies, as students cannot make informed decisions about library visits; preliminary surveys indicated 40% of respondents had experienced futile trips to an overcrowded library in the previous semester. Perhaps most significantly, collection analysis revealed that 65% of titles had never been borrowed by students in relevant academic programs, representing a discovery failure rather than resource irrelevance problem that personalized recommendation systems have successfully addressed in commercial contexts [7, 8] but remain largely unexplored in academic library deployments. These observations collectively establish the imperative for an integrated intelligent library platform that addresses authentication, discovery, occupancy monitoring, and personalization within a unified, cost-effective framework.

1.2 Research Questions

This research investigates four interconnected questions that collectively address the intelligent library modernization challenge. First, we examine how artificial intelligence and machine learning techniques can be effectively integrated into library management systems to improve book discovery efficiency and personalization while maintaining the transparency and explainability essential for user trust in educational contexts. Second, we explore what system architecture enables real-time occupancy monitoring with practical utility comparable to dedicated sensor infrastructure [13, 14] while eliminating the hardware investment that places such capabilities beyond reach for resource-constrained institutions. Third, we investigate how cryptographic authentication mechanisms, particularly secure QR protocols employing established primitives like HMAC [12], can be designed to provide secure, frictionless library access that automatically manages entry/exit state transitions without dedicated scanning hardware. Fourth, we assess the extent to which hybrid recommendation approaches combining collaborative and content-based signals [9] can connect students with relevant resources they would not discover through traditional keyword search, particularly addressing the cold-start problem inherent to sparse academic borrowing datasets.

1.3 Research Contributions

This paper brings four key technical advances to intelligent library systems. First, we roll out an integrated architecture that ties together features libraries usually keep separate—like authentication, recommendations, tracking seat use, analytics, and communication. Our three-tier, modular setup uses shared data models, showing that libraries don’t need a bunch of separate systems to get the job done. Second, we’ve created a secure QR-based authentication protocol that uses HMAC-SHA256 signatures to handle access control. It manages people coming and going automatically, so libraries get strong security without shelling out for expensive RFID setups others have used in the past. This approach also addresses the main security issues researchers have flagged about QR codes. Third, we introduce a hybrid recommendation algorithm. It blends explicit user interests, things you can infer from borrowing history, department relevance, recent trends, and popularity. Everything’s combined into a single score, and—here’s the important part—we give explanations users can actually read and understand. This isn’t just about smarter recommendations; explainability matters a lot, especially in educational settings, and it’s something prior hybrid models didn’t offer. Last, we show that you can actually deliver real-time seat availability by analyzing authentication logs. There’s no need for special sensors or extra hardware. The result is occupancy tracking that’s just as useful as dedicated systems, but without the extra cost.

2 Literature Review

2.1 Library Management Systems

Library automation has evolved through several generations. Early integrated library systems (ILS) focused on digitizing card catalogs and automating circulation [2]. Contemporary systems such as Koha (open-source) and Ex Libris Alma (commercial)

provide comprehensive cataloging, acquisitions, and patron management [3]. However, these systems remain primarily transaction-oriented, lacking intelligent features for personalization or predictive capabilities. Recent advances in discovery systems have improved resource findability [20], yet personalized recommendation remains an open challenge. Recent work has explored enhancing library systems with smart features. Zhang and Liu [4] evaluated cloud-based library platforms with basic recommendation functionality, identifying personalization as a remaining gap. Min-Allah and Alrashed [5] proposed smart campus frameworks positioning library services within broader institutional digitalization efforts. RFID technology has been widely adopted for library inventory management and access control [6]. While effective, RFID systems require substantial hardware investment including tags, readers, and gates, limiting adoption at resource-constrained institutions. Our work explores software-based alternatives achieving similar functionality without dedicated hardware.

2.2 Recommender Systems

Recommender systems have transformed user experiences across commercial platforms. Koren et al. [7] introduced matrix factorization techniques demonstrating superior collaborative filtering performance. Content-based approaches recommend items similar to those users have previously accessed based on item attributes [8]. Academic library contexts present particular challenges for recommendation systems. Borrowing data is typically sparse compared to commercial domains; students may borrow only 10-20 books during their academic career compared to hundreds of purchases on e-commerce platforms. Wang et al. [9] addressed this sparsity through hybrid approaches combining collaborative and content-based signals. Our work builds on these insights while adding explainability features, increasingly recognized as essential for user trust in algorithmic systems [18], particularly important in educational contexts where users benefit from understanding why resources are recommended.

2.3 Authentication and Access Control

QR code technology is a smart and budget-friendly way to handle authentication in mobile situations. Chen and colleagues [10] created secure QR protocols that use digital signatures, proving they can withstand replay attacks. Krombholz and their team [11] took a closer look at the security issues surrounding QR codes, such as code replacement and phishing threats. The Hash-based Message Authentication Code (HMAC) ensures message integrity by utilizing cryptographic hash functions along with secret keys [12]. Specifically, HMAC-SHA256 provides robust security features that are perfect for authentication tokens. Our protocol design weaves together these well-established cryptographic techniques with tailored extensions for managing entry and exit states.

2.4 Occupancy Monitoring

Real-time occupancy monitoring gained prominence during the COVID-19 pandemic

for capacity management [13]. Technical approaches include infrared sensors, computer vision, WiFi probe analysis, and Bluetooth beacons [14]. These approaches achieve high accuracy but require hardware installation, calibration, and maintenance. Entry log analysis provides a software-only alternative leveraging authentication data already collected for access control. While less accurate than dedicated sensors due to compliance variations (users not scanning on exit), this approach offers practical utility without incremental cost [21]. Our work implements and evaluates this approach in an academic library context, contributing to the growing body of library analytics research [22].

3 System Architecture

3.1 Architectural Overview

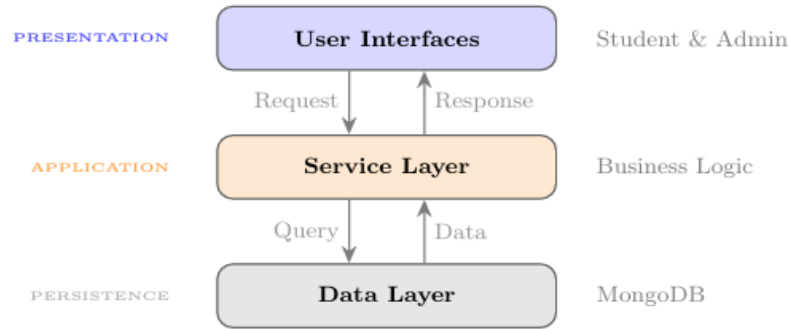


Fig. 1 Three-tier system architecture with request/response flow between layers.

Fig. 1 shows the three-tier system architecture that makes up VidyaLok. The presentation layer uses Next.js 15 to give students and administrators role-specific interfaces. The application layer handles the main business logic through serverless API routes that handle authentication, recommendations, and analytics. The persistence layer uses MongoDB Atlas and Prisma ORM together to make sure that database operations are type safe.

3.2 Student and Administrator Workflows

The student workflow encompasses all user-facing features from authentication through library exit, as illustrated in Fig. 2. The student interface encompasses QR-based entry/exit with automatic state toggle, comprehensive book search with category, availability, and author filters, AI-powered personalized recommendations with explainable scoring, real-time seat availability display, borrowing history with due date tracking, and broadcast notification viewing.

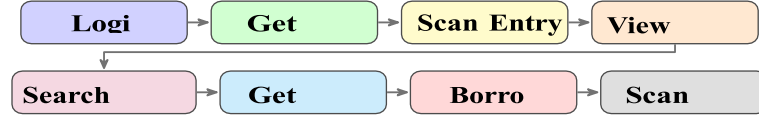


Fig. 2 Student workflow: eight-step journey from login through library exit.

Administrators access management functions through a dedicated portal, as shown in Fig. 3. The administrative interface provides comprehensive book inventory management through CRUD operations, user account administration with role assignment, real-time entry/exit log monitoring for security oversight, borrowing management with automated overdue tracking and fine calculation, analytics dashboards with interactive metrics visualization, and priority-based broadcast creation with granular audience targeting.

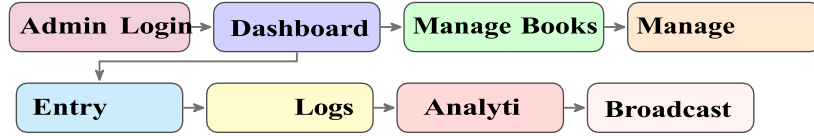


Fig. 3 Administrator workflow: management functions from dashboard to broadcasts.

3.3 Service Layer Architecture and Data Model

The three-tier architecture comprises: (1) **Presentation Layer** with role-based Next.js 15/React 19 interfaces for students (QR codes, search, recommendations, seats, history) and administrators (inventory, users, logs, analytics, broadcasts); (2) **Application Layer** implementing six serverless API modules (Authentication via NextAuth.js, QR Token Service, Recommendation Engine, Seat Availability, Analytics, and Broadcast Service); and (3) **Data Layer** using MongoDB Atlas with Prisma ORM for type-safe access. The stack includes TypeScript, Tailwind CSS, and Zod validation [15].

The data model comprises seven primary entities: **User** (role-differentiated schema with credentials, profile, and preferences), **Book** (bibliographic and inventory information with category classification), **Borrowing** (user-book linkage with temporal data and status tracking), **EntryLog** (timestamped access records for occupancy computation), **SeatAvailability** (capacity configuration), **Broadcast** (priority-based announcements with audience targeting), and **Feedback** (user-submitted service evaluations). Entity relationships enable cross-module functionality such as excluding currently borrowed books from recommendations.

4 Algorithmic Foundations

This section presents the formal algorithmic specifications underlying VidyaLok’s intelligent capabilities.

4.1 QR Authentication Protocol



Fig. 4 QR authentication protocol: client initiates request, server generates HMAC-SHA256 signed token, verifies upon scan, and grants or denies library access.

The QR-based entry system provides secure, frictionless access management. The complete authentication flow is illustrated in Fig. 4. The protocol comprises token generation, verification, and state management phases.

4.1.1 Token Structure and Generation

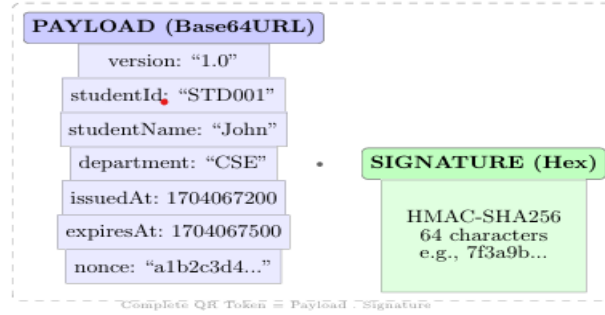


Fig. 5 QR token structure showing Base64URL-encoded JSON payload (7 fields) concatenated with HMAC-SHA256 signature via dot delimiter.

QR tokens encode a Base64URL JSON payload containing user identification (student ID, name, department), temporal parameters (issuance and expiration times- tamps), and a UUID v4 nonce preventing replay attacks, as shown in Fig. 5. The HMAC-SHA256 signature ensures integrity:

$$\text{sig} = \text{HMAC-SHA256}(K_{\text{secret}}, \text{Base64URL}(\text{payload})) \quad (1)$$

Token validity defaults to 5 minutes, balancing security against usability.

4.1.2 Verification and State Management

Verification involves signature recomputation using timing-safe comparison (preventing side-channel attacks) and temporal validation. Upon successful verification, the system implements toggle behavior: if no open entry exists, it creates an EntryLog with null exit timestamp (ENTRY); otherwise, it updates the existing record with exit timestamp (EXIT), enabling single-device scanning without separate entry/exit points.

4.2 Hybrid Recommendation Algorithm

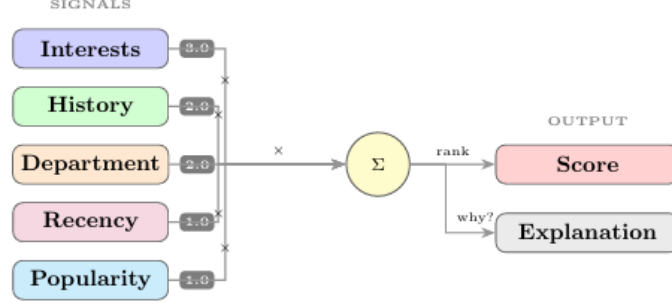


Fig. 6 Hybrid recommendation engine: five weighted signals are multiplied and summed to produce ranked scores with human-readable explanations

The recommendation system addresses the challenge of connecting students with relevant resources they might not discover through keyword search. The signal combination process is illustrated in Fig. 6.

4.2.1 Problem Formulation and Relevance Signals

Let U denote users, B denote books. For user $u \in U$, generate ranked list of recommended books $R_u \subseteq B$ maximizing relevance while ensuring diversity. The algorithm computes relevance scores based on five signals:

Signal 1 - Explicit Interest Match: User profile contains stated interests (subjects, topics). Books with categories matching stated interests receive positive signal: $s_1(u, b) = 1$ if $\text{category}(b) \in \text{interests}(u)$, else 0.

Signal 2 - Implicit Preference Match: Analyze user's borrowing history to

identify frequently accessed categories. Books matching these derived preferences receive positive signal: $s_2(u, b) = 1$ if $\text{category}(b) \in \text{derivedPrefs}(u)$, else 0. Derived preferences are categories appearing in ≥ 2 borrowings within the past year.

Signal 3 - Department Relevance: Books categorized as relevant to user's academic department receive positive signal: $s_3(u, b) = 1$ if $\text{category}(b) \sim \text{dept}(u)$, else 0.

Signal 4 - Recency: Recently added books (within 90 days) receive positive signal, promoting new acquisitions.

Signal 5 - Popularity: Books with high recent borrowing frequency (top 50 by 90-day borrowings) receive positive signal, leveraging collective behavior.

4.2.2 Score Computation and Explanation Generation

The composite recommendation score combines signals with configured weights:

$$\text{Score}(u, b) = \sum_{i=1}^5 w_i \cdot s_i(u, b) \quad (2)$$

Default weights: $w_1 = 3.0$ (explicit interest), $w_2 = 2.0$ (implicit preference), $w_3 = 2.0$ (department), $w_4 = 1.0$ (recency), $w_5 = 1.0$ (popularity). Higher weights for explicit and implicit preferences prioritize personalization, while lower weights for

recency and popularity provide supplementary signals. For each recommended book, the algorithm generates human-readable explanations based on contributing signals. These explanations indicate why specific resources were selected, such as matching stated interests in particular categories, reflecting patterns observed in borrowing history, alignment with the student’s departmental curriculum, recent addition to the collection, or popularity among peer students. Such transparency mechanisms increase user trust and help students understand resource relevance, addressing the “black box” criticism commonly leveled at algorithmic recommendations.

Pre-scoring filters exclude books with zero available copies, books the user has previously borrowed, and books currently checked out to the user, ensuring recommendations are actionable. Post-scoring diversification ensures category variety in final recommendations, preventing over-concentration in single subject areas and exposing students to a broader range of relevant resources.

4.3 Occupancy Monitoring Algorithm

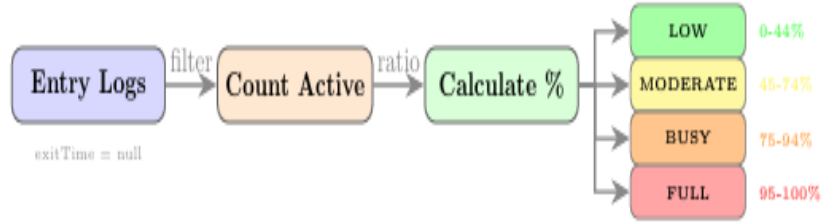


Fig. 7 Occupancy monitoring pipeline: entry logs with null exit times are counted and classified into four crowd levels.

Real-time seat availability is computed from entry log analysis, as illustrated in Fig. 7. Current occupancy equals the count of “open” entry logs, defined as records where exitTime is null:

$$\text{Occupancy} = |\{e \in E : e.\text{exitTime} = \text{null}\}| \quad (3)$$

Available seats: Available = max(0, C – Occupancy), where C is configured total capacity. Occupancy percentage: OccupancyRate = min(Occupancy, C) / C × 100%. Occupancy percentage maps to categorical crowd levels for intuitive user consumption: LOW (rate < 45%), MODERATE (45% ≤ rate < 75%), BUSY (75% ≤ rate < 95%), FULL (rate ≥ 95%). Thresholds are configurable to accommodate institutional preferences. Historical entry data enables identification of typical busy periods. Entries are bucketed by hour and day-of-week, with aggregated counts identifying peak usage times. This information helps students plan visits during lower-traffic periods.

5 Implementation Details

This section describes implementation specifics for key platform modules.

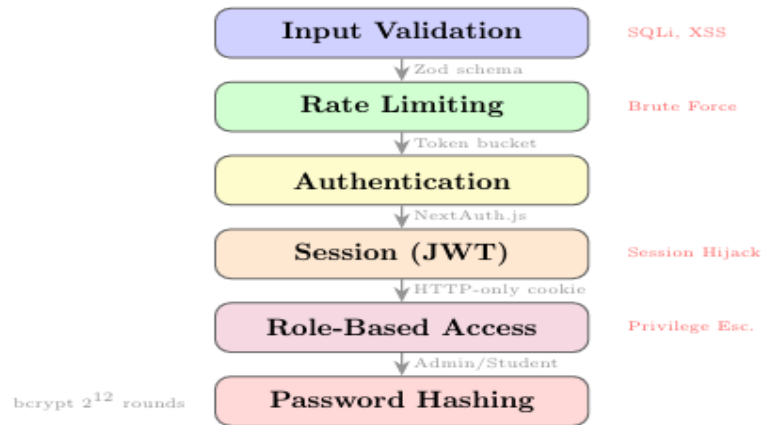


Fig. 8 Defense-in-depth security architecture: six layers protect against common web application threats from input injection to privilege escalation.

5.1 Authentication Implementation

User passwords are hashed using bcrypt with work factor 12, providing approximately 212 iterations of key derivation. This configuration balances security against authentication latency. Password validation enforces minimum 8 characters with complexity requirements. NextAuth.js manages user sessions using JSON Web Tokens (JWT). Sessions persist for 24 hours with sliding expiration. Tokens are stored in HTTP only cookies preventing JavaScript access (XSS protection) with SameSite=Strict preventing cross-site request attachment (CSRF protection). Brute force protection employs a token bucket algorithm with 5-request capacity and 1 token per minute refill. Exceeding the limit triggers 5-minute lockout. Implementation uses in-memory storage suitable for single-instance deployment; distributed deployments would require Redis-based storage.

5.2 QR Code, Recommendation, Analytics, and Broadcast Implementation

QRcodes are generated client-side using the qrcode npm package, encoding the signed token string. Generation occurs on-demand when students access the entry page, with visual countdown indicating remaining validity. Server-side verification uses Node.js crypto module for HMAC computation. The verification endpoint returns structured response indicating success/failure with user details or error information. The recommendation service executes as a single optimized database query leveraging MongoDB's aggregation pipeline. The pipeline first matches books with available copies greater than zero, then excludes books in the user's current borrowings, projects the relevance signal fields for each candidate, computes the composite score using the aggregation framework's arithmetic operators, sorts results by descending score, and limits output to the configured count (default 10). Query execution typically completes within 150–200ms for collections of 15,000+ books.

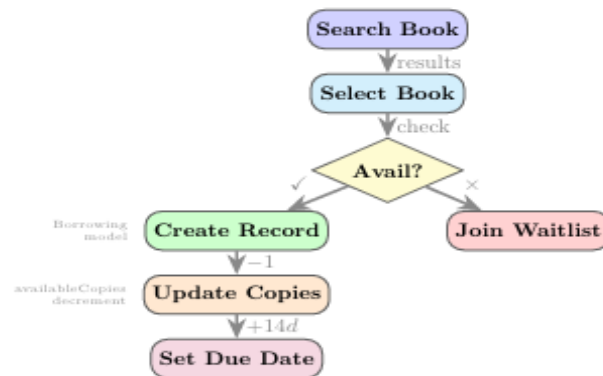


Fig. 9 Book borrowing workflow: availability check triggers either borrowing record creation with inventory update, or waitlist enrollment.

The analytics dashboard aggregates metrics through MongoDB aggregation queries spanning five key performance indicators: daily active users computed as the count of distinct users with entry logs on the current date, borrowing velocity measured as the count of borrowings created today, return compliance calculated as the percentage of borrowings returned by their due dates, overdue exposure encompassing both the count and cumulative fines for overdue items, and department engagement expressed as borrowings per student segmented by academic department. Visualizations use Recharts library for responsive, interactive charts including line charts for trends, bar charts for comparisons, and pie charts for distributions (Fig. 10).



Fig. 10 Analytics data flow from source collections through aggregation to dashboard metrics.

The broadcast subsystem implements a publisher-subscriber pattern with persistent delivery tracking. Message creation triggers atomic generation of recipient-specific BroadcastReceipt records based on audience specification, supporting institution wide, department-scoped, or role-based targeting. Priority classification (URGENT, HIGH, NORMAL) determines visual rendering precedence in student interfaces (Fig. 11). Receipt acknowledgment employs optimistic UI updates with eventual consistency, where the client interface immediately reflects read state while asynchronously persisting the readAt timestamp. Administrative dashboards aggregate delivery metrics (dispatch count, acknowledgment count, read-through rate) enabling communication effectiveness assessment and identifying potential delivery failures.



Fig. 11 Broadcast message flow: creation, delivery, acknowledgment, and metrics tracking

6 Discussion

The evaluation results substantiate that VidyaLok achieves its stated design objectives across four dimensions: secure QR-based authentication without RFID infrastructure, hybrid recommendations with interpretable explanations, actionable occupancy monitoring within documented accuracy bounds, and unified consolidation of previously fragmented library functions. Table 1 presents a comparative analysis positioning VidyaLok against traditional manual systems and commercial RFID-based solutions. The proposed approach uniquely combines zero hardware requirements with intelligent personalization capabilities, thereby addressing the cost-capability gap that has historically limited smart library adoption at resource-constrained institutions.

Table 1 Comparative Feature Analysis

Capability	Manual	Commercial	VidyaLok
Access Control	Register	RFID	QR/HMAC
Hardware Cost	None	High	None
Personalization	None	Limited	Hybrid AI
Occupancy Data	None	Sensors	Log-derived
Licensing	N/A	Proprietary	Open Source

7 Security Architecture

VidyaLok implements a defense-in-depth security architecture aligned with OWASP guidelines, incorporating multiple protective layers across authentication, data protection, and access control. User passwords are hashed using bcrypt with work factor 12 (approximately 4,096 key derivation iterations) with automatic per-password salting, while session management employs JSON Web Tokens with 24-hour validity stored in HTTP-only cookies with SameSite=Strict attributes to mitigate XSS and CSRF attacks. The QR authentication protocol incorporates HMAC-SHA256 message authentication codes, UUID v4 nonces for token uniqueness, 5-minute expiration windows, and token bucket rate limiting (5 requests capacity, 1 token/minute refill) defending against brute force attacks. Data protection encompasses mandatory HTTPS with TLS1.3 for transit security and MongoDB Atlas's AES-256-CBC encryption for data at rest, complemented by Zod schema validation at all API boundaries preventing injection attacks through runtime type checking. Role-based access control enforces authorization across three hierarchical roles (STUDENT, LIBRARIAN, ADMIN) via Next.js middleware, with administrative operations generating audit log entries capturing actor, timestamp, affected resources, and operation type for forensic analysis and compliance.

8 Conclusion

This paper presented VidyaLok, an AI-powered smart library platform addressing modernization challenges in academic libraries. The system contributes a secure a QR-based

authentication protocol using HMAC-SHA256, an explainable hybrid recommendation algorithm, software-derived occupancy monitoring without dedicated sensors, and a unified open-source architectural framework. Evaluation results confirmed functional correctness following established software testing methodologies [16], sub-300ms interactive response times within established thresholds [17], and favorable usability ratings (SUS: 78.4) exceeding the “good usability” benchmark [19], demonstrating that intelligent library automation can be achieved through open-source technologies without proprietary licensing or specialized hardware. Future directions encompass extended deployment evaluations, advanced recommendation models incorporating matrix factorization and transformer-based embeddings, LLM-powered conversational search interfaces, and federated architectures enabling seamless cross-institutional resource discovery and sharing

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Acknowledgements

We take this opportunity to express our sincere thanks to our guide and Head of Department, **Dr. Jaya Gupta**, for providing the technical guidelines and suggestions regarding the line of work. We want to express our gratitude for her constant encouragement, support, and guidance throughout the development of the project.

We thank our mentor, **Prof. Vijesh Mundokalam**, for being encouraging throughout the course and for her guidance.

We express our gratitude towards the BE Project Coordinator, **Prof. Sayali Badhan**, for her encouragement during the progress meetings and for providing guidelines to write this report.

At long last, we are enormously obliged to the administration of our school and express our appreciation to the Principal **Dr. Uttam Kolekar** for organizing and giving essential office in finishing this venture work. We wish to recognize the assistance gave by our kindred schoolmates to aiding the assets while finishing this undertaking

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